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Discussion Paper

Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment*

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Abstract

Despite an increasing reliance on fully-automated algorithmic decision-making in our day-to-day lives, humans still make consequential decisions. While the existing literature focuses on the bias and fairness of algorithmic recommendations, an overlooked question is whether they improve human decisions. We develop a general statistical methodology for experimentally evaluating the causal impacts of algorithmic recommendations on human decisions. We also examine whether algorithmic recommendations improve the fairness of human decisions and derive the optimal decision rules under various settings. We apply the proposed methodology to the first-ever randomized controlled trial that evaluates the pretrial Public Safety Assessment in the United States criminal justice system. Our analysis of the preliminary data shows that providing the PSA to the judge has little overall impact on the judge's decisions and subsequent arrestee behaviour.

Keywords: algorithmic fairness, causal inference, principal stratification, randomized experiments, recommendation systems, sensitivity analysis

1 Introduction

A growing body of literature has suggested the potential superiority of algorithmic decision-making over purely human choices across a variety of tasks (e.g., Hansen & Hasan, 2015; He et al., 2015). Although some of this evidence is decades old (e.g., Dawes et al., 1989), it has recently gained significant public attention for the spectacular defeats of humanity's best in cerebral games (e.g., Silver et al., 2018). Yet, even in contexts where research has warned of human frailties, we humans still make many consequential decisions for a variety of reasons including the preservation of human agency and accountability.

The desire for a human decision-maker as well as the precision and efficiency of algorithms have led to the adoption of hybrid systems involving both. By far the most popular system uses algorithmic recommendations to inform human decision-making. Such algorithm-assisted human decision-making has been deployed in many aspects of our daily lives, including medicine, hiring, credit lending, investment decisions, and online shopping. And of particular interest, algorithmic

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recommendations are increasingly of use in the realm of evidence-based public policy-making. A prominent example, studied in this paper, is the use of risk assessment instruments in the criminal justice system that are designed to improve incarceration rulings and other decisions made by judges.

While there exists a fast-growing literature in computer science that studies the bias and fairness of algorithms (see Chouldechova & Roth, 2020, for a review and many references therein), an overlooked question is whether such algorithms help humans make better decisions (see e.g., Green & Chen, 2019, for an exception). In this paper, we develop a general methodological framework for experimentally evaluating the impacts of algorithmic recommendations on human decision-making. We conducted the first-ever real-world field experiment by providing, for a randomly selected cases, information from a system consisting of Public Safety Assessment (PSA) risk scores and a recommendation from a Decision-Making Framework (DMF) to a judge who makes an initial release decision. We evaluate whether the PSA-DMF system (which for brevity we refer to as the PSA hereafter) helps judges achieve their goal of preventing arrestees from committing a new crime or failing to appear in court while avoiding an unnecessarily harsh decision.

Using the concept of principal stratification from the causal inference literature (e.g., Ding & Lu, 2017; Frangakis & Rubin, 2002), we propose the evaluation quantities of interest, identification assumptions, and estimation strategies. We also develop sensitivity analyses to assess the robustness of empirical findings to the potential violation of a key identification assumption (see also Hirano et al., 2000; Jiang et al., 2016; Mattei et al., 2013; Schwartz et al., 2011). In addition, we examine whether algorithmic recommendations improve the fairness of human decisions, using the concept of principal fairness that, unlike other fairness criteria, accounts for how the decision in question affects individuals (Imai & Jiang, 2023). Finally, we consider how the data from an experimental evaluation can be used to inform an optimal decision rule and assess the optimality of algorithmic recommendations and human decisions (see Ben-Michael et al., 2021, for a methodological framework for learning an optimal algorithmic recommendation). Although we describe and apply the proposed methodology in the context of evaluating the PSA, it is directly applicable or extendable to many other settings of algorithm-assisted human decision-making.

The use of risk assessment scores, which serves as the main application of the current paper, has played a prominent role in the literature on algorithmic fairness since the controversy over the potential racial bias of COMPAS risk assessment score used in the United States (US) criminal justice system (see e.g., Angwin et al., 2016; Dieterich et al., 2016; Dressel & Farid, 2018; Flores et al., 2016). With few exceptions, however, much of this debate focused on the accuracy and fairness properties of risk assessment scores itself rather than how they affect judges' decisions (see e.g., Berk et al., 2018; Kleinberg et al., 2018; Rudin et al., 2020, and references therein). Even studies that directly estimate the impacts of risk assessment scores on judges' decisions are based on either observational data or hypothetical vignettes in surveys (e.g., Albright, 2019; Berk, 2017; Garrett & Monahan, 2020; Green & Chen, 2019; Miller & Maloney, 2013; Skeem et al., 2020; Stevenson, 2018; Stevenson & Doleac, 2021).

We contribute to this literature by demonstrating how to evaluate the use of risk assessment scores experimentally when humans are ultimate decision-makers. To the best of our knowledge, this is the first real-world randomized controlled trial (RCT) that evaluates the impacts of algorithmic risk assessment scores on judges' decisions in the criminal justice system (see also the Manhattan Bail Project and Philadelphia Bail Experiment that evaluated the effects of bail guidelines on judges' decisions several decades ago Ares et al., 1963; Goldkamp & Gottfredson, 1984, 1985). Using the concept of principal stratification from causal inference literature, the proposed methodology allows us to evaluate the effects of the PSA on judges' decisions separately for the subgroups of arrestees with different levels of risks.

Based on the preliminary data from our experiment (complete data will not be available for some time), we find that the provision of the PSA has little overall impact on the judge's decisions across three outcomes we examine: failure to appear (FTA), new criminal activity (NCA), and new violent criminal activity (NVCA). Our analysis, however, provides some suggestive evidence that the PSA may make the judge's decisions more lenient for female arrestees regardless of their risk levels, while it encourages the judge to make stricter decisions for male arrestees who are deemed to be risky. In terms of fairness, the PSA appears to increase an existing gender difference while having no substantial impact on any racial differences in judges' decisions. Finally, we use the

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experimental data to learn about the optimal decision rule that minimizes the prevalence of negative outcomes (FTA, NCA, and NVCA) while avoiding unnecessarily harsh decisions. Our analysis suggests that the PSA's recommendations may be unnecessarily severe unless a jurisdiction considers the costs of FTA, NCA, and NVCA to be sufficiently high. This might suggest that incarceration decisions themselves, whether PSA-informed or otherwise, are also unnecessarily severe.

2 Experimental evaluation of pretrial public safety assessment

In this section, we briefly describe our field experiment after providing some background about the use of the PSA in the US criminal justice system. Additional details about our experiment are given in Greiner et al. (2020).

2.1 Background

The US criminal justice apparatus consists of thousands of diverse systems. Some are similar in the decision points they feature, as an individual suspected of a crime travels from the investigation to sentencing. Common decision points include whether to stop and frisk an individual in a public place, whether to arrest or issue a citation to an individual suspect of committing a crime, whether to release the arrestee while they await the disposition of any charges against them (the subject of this paper), what charge(s) to be filed against the individual, whether to find the defendant guilty of those charges, and what sentence to impose on a defendant found guilty.

At present, human judges make all these decisions. In theory, algorithms could inform any of them and could even make some of these decisions without human involvement. To date, algorithmic outputs have appeared most frequently in two settings: (i) at the 'first appearance' hearing, during which a judge decides whether to release an arrestee pending disposition of any criminal charges and (ii) at sentencing, in which the judge imposes a punishment on a defendant found guilty. The first of these two motivates the present paper, but the proposed methodology is applicable or extendable to other settings.

We describe a typical first appearance hearing. The key decision the judge must make at a first appearance hearing is whether to release the arrestee pending disposition of any criminal charges and, if the arrestee is to be released, what conditions to impose. Almost all jurisdictions allow the judge to release the arrestee with only a promise to reappear at subsequent court dates. In addition, because arrestees have not yet been adjudicated guilty of any charge at the time of a pretrial hearing, there exists a consensus that pretrial incarceration is to be avoided unless the risks associated with release are sufficiently high.

Judges deciding whether to release arrestees ordinarily consider two risk factors among a variety of other concerns; the risk that the arrestee will fail to appear at subsequent court dates, and the risk that the arrestee will engage in new criminal activity (NCA) before the case is resolved (e.g., 18 U.S.C. § 3142(e)(1)). Jurisdiction laws vary regarding how these two risks are to be weighed. Some jurisdictions direct judges to consider both simultaneously along with other factors (e.g., Ariz. Const. art. II, § 22, Iowa Code § 811.2(1)(a)), while others focus on only FTA risk (e.g., N.Y. Crim. Proc. Law § 510.30(2)(a)). Despite these variations, NCA and FTA are constant and prominent in the debate over the first appearance decisions.

Concerns about the consequential nature of the first appearance decision have led to the development of the PSA, which is ordinarily offered as an input to first appearance judges. Predisposition risk assessment instruments take various forms, but most focus on classifying arrestees according to FTA and NCA risks. They are generally constructed by fitting a statistical model to a training dataset based on the past first appearance hearings and the subsequent incidences (or lack thereof) of FTA and NCA. The hope is that providing such instruments will improve the assessment of FTA and NCA risks and thereby lead to better decisions. The goal of this paper is to develop a general methodological framework for evaluating the impact of providing the PSA to judges at first appearance hearings using an RCT, to which we now turn.

2.2 The experiment

We conducted a field RCT in Dane county, Wisconsin, to evaluate the impacts of PSA provision on judges' decisions. The PSA consists of three scores—two six-point scores separately summarizing

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FTA and NCA risks as well as a binary score for the risk of NVCA. These scores are based on the weighted indices of nine factors drawn from criminal history information, primarily prior convictions and FTA, and a single demographic factor, age. Notably, gender and race are not used to compute the PSA. The weights are calculated using past data. A Decision-Making Framework (DMF) combines information from the three PSA scores with other considerations to produce an overall recommendation to the judge, which the judge may accept or modify or ignore as they see fit. The details about the construction of the PSA and other relevant information are available at https://advancingpretrial.org/psa/factors/

The field operation was straightforward. In this county, a court employee assigned each matter a case number sequentially as it entered the system. No one but this clerk was aware of the pending matter numbers, so manipulation of the number by charging assistant district attorneys was not possible. Employees of the Clerk's office scanned online record systems to calculate the PSA for all cases. If the last digit of the case number was even, these employees made the PSA (specifically, a printout of the PSA scores, the DMF recommendation, and the supporting criminal history and age information) available to the judge. Otherwise, no PSA was made available. Thus, the provision of the PSA to judges was essentially randomized. Indeed, the comparison of the observed covariate distributions suggests that this scheme produced groups comparable on background variables (Greiner et al., 2020).

The judge presiding over the first appearance hearing by law was to consider the risk of FTA and NCA, along with other factors, including ties to the community as prescribed by statute. The judge could order the arrestee released with or without bail of varying amount. The judge could also condition release on compliance with certain conditions such as monitoring, but for the sake of simplicity, we focus on bail decisions and ignore other conditions in this paper.

When making decisions, the judge also had information other than the PSA and its inputs. In all cases, the judge had a copy of an affidavit sworn to by a police officer recounting the circumstances of the incident that led to the arrest. The defense attorney sometimes informed the judge of the following regarding the arrestee's connections to the community: length of time lived there, employment there, and family living there. When available, this information ordinarily stemmed from an arrestee interview conducted earlier by a paralegal. The assistant district attorney sometimes provided additional information regarding the circumstances of the arrest or criminal history. Given the lack of access to this additional information, we develop a sensitivity analysis to address a potential unobserved confounding bias.

2.3 The data

The field operation design called for approximately a 30-month treatment assignment period (from the middle of 2017 until the end of 2019) followed by the collection of data on FTA, NCA, NVCA, and other outcomes for a period of two years after randomization. At the time of this writing, we have outcome data from a 12-month follow-up of each first appearance event that occurred in the first 12 months of randomization. The 30-month randomization period has expired, and we will report the results of our comprehensive analysis of a full dataset in the future. Furthermore, although some arrestees had multiple cases during the study period, this paper focuses only on the first of the first appearance hearings for any individual arrestee. This leads to a total of 1891 cases for our analysis, of which 40.0% (38.8%) are white male arrestees and 13.0 are white female arrestees (non-white male and female arrestees account for 38.8% and 8.1%, respectively).

Based on the empirical distribution of bail amounts and expert's opinion, we categorize the judge's decisions into three ordinal categories: signature bond, small cash bond (less than $1,000), and large cash bond (greater than or equal to $1,000). A signature bond requires an arrestee to sign a promise to return to the court for trial but does not require any payment to be released. Cash bonds require an arrestee to deposit money with the court to obtain release. Table 1 summarizes the joint distribution of treatment assignment (PSA provision), the judge's decisions (three ordinal categories), and three binary outcomes. We observe that in about three quarters of cases, the judge imposed signature bonds, while in the remaining cases, the judge imposed bail. For the outcome variables, slightly less than 30% of arrestees commit FTA or NCA whereas the proportion of those who commit NVCA is only about 6%.

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Table 1. The Joint Distribution of Treatment Assignment, Judge's Decisions, and Outcomes. The table shows the number of cases in each category with the corresponding percentage in parentheses. Only about 20% of all arrestees are female. Few cases result in NVCA (new violent criminal activity), while FTA (failure to appear in court) and NCA (new criminal activity) occur in slightly above 25% each. A majority of decisions are signature bonds rather than cash bonds.

	no PSA (Control group)			PSA (Treatment group)			
	Signature	Cash bond		Signature	Cash bond		
	bond	≤$1,000	>$1,000	bond	≤$1,000	>$1,000	Total (%)
Non-white Female	64	11	6	67	6	0	154
	(3.4)	(0.6)	(0.3)	(3.5)	(0.3)	(0.0)	(8.1)
White Female	91	17	7	104	17	10	246
	(4.8)	(0.9)	(0.4)	(5.5)	(0.9)	(0.5)	(13.0)
Non-white Male	261	56	49	258	53	57	734
	(13.8)	(3.0)	(2.6)	(13.6)	(2.8)	(3.0)	(38.8)
White Male	289	48	44	276	54	46	757
	(15.3)	(2.5)	(2.3)	(14.6)	(2.9)	(2.4)	(40.0)
FTA committed	218	42	16	221	45	16	558
	(11.5)	(2.2)	(0.8)	(11.7)	(2.4)	(0.8)	(29.4)
not committed	487	90	90	484	85	97	1333
	(25.8)	(4.8)	(4.8)	(25.6)	(4.5)	(5.1)	(70.6)
NCA committed	211	39	14	202	40	17	523
	(11.2)	(2.1)	(0.7)	(10.7)	(2.1)	(0.9)	(27.7)
not committed	494	93	92	503	90	96	1368
	(26.1)	(4.9)	(4.9)	(26.6)	(4.8)	(5.1)	(72.4)
NVCA committed	36	10	3	44	10	6	109
	(1.9)	(0.5)	(0.2)	(2.3)	(0.5)	(0.3)	(5.7)
not committed	669	122	103	661	120	107	1782
	(35.4)	(6.5)	(5.4)	(35.0)	(6.3)	(5.7)	(94.3)
Total	705	132	106	705	130	113	1891
	(37.3)	(7.0)	(5.6)	(37.3)	(6.9)	(6.0)	(100)

2.4 The overall impact of PSA provision on judge's decisions

Figure 1 presents the distribution of the judge's decisions given each of the PSA scores among the cases in the treatment (top panel) and control (bottom panel) groups. The overall difference in the conditional distribution between the two groups is small though there are some differences in some subgroups (see Online Supplementary Material, Appendix S1). The PSA scores for FTA and NCA are ordinal, ranging from 1 (safest) to 6 (riskiest), whereas the PSA score for NVCA is binary, 0 (safe) and 1 (risky). We also plot the DMF recommendation, which aggregates these three PSA scores as well as other information such as types of charges. The DMF recommendation has four categories (signature bond, modest cash bond, moderate cash bond, and cash bond with maximum conditions), but we dichotomize it into signature or cash bond given its skewed empirical distribution.

In general, we observe a positive association between the PSA scores and judge's decisions, implying that a higher PSA score is associated with a harsher decision. We also find that for FTA and NCA, the most likely scores are in the medium range, while the vast majority of NVCA cases were classified as no elevated risk. For NCA and FTA, the judge's decisions varied little when the PSA score took a value in the lower range. For the DMF recommendation, the judge is far more likely to give a signature bond for the cases that are actually recommended for a signature bond.

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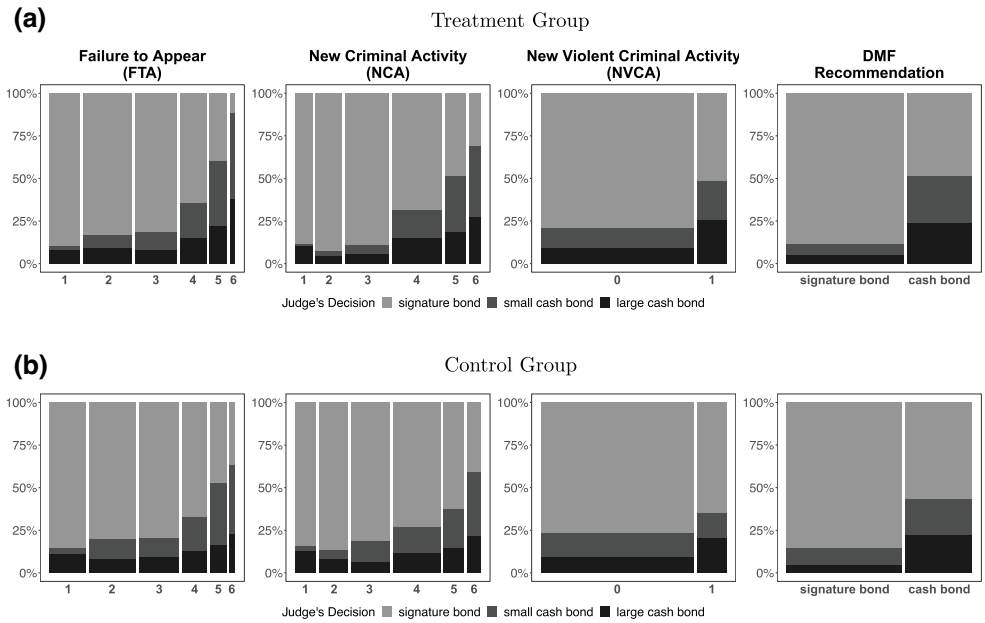

Figure 1. The distribution of the Judge's Decisions given the Pretrial Public Safety Assessment (PSA) among the Cases in the Treatment (Top Panel) and Control (Bottom Panel) Groups. There are three PSA scores, two of which are ordinal—FTA and NCA—while the other is dichotomous—NVCA. The judge's decision is coded as a three-category ordinal variable based on the type and amount of bail: a signature bond, a small cash bond (less than $1,000), and a large cash bond (greater than or equal to $1,000). The DMF recommendation is presented as a binary variable: signature or cash bond. The width of each bar is proportional to the number of cases for each value of the corresponding PSA score. There exists a positive correlation between PSA scores and the severity of the judge's decisions in both treatment and control groups. (a) Treatment group. (b) Control group.

Figure 2 presents the estimated average causal effect of PSA provision on the judge's decisions (left plot) and three outcomes of interest (right plot). We use the difference-in-means estimator and display the 95% confidence intervals as well as the point estimates. We do not compute separate estimates for white females and non-white females because we have too few female arrestees (see Table 1). The results imply that PSA provision, on average, has little effect on the judge's decisions. In addition, the average effects of PSA provision on the three outcomes are also largely ambiguous although there is suggestive evidence that it may slightly increase NVCA among female arrestees. In Online Supplementary Material, Appendix S2.1, we also explore the average causal effects of PSA provision across different age groups. We find some suggestive causal effects for the group of 29–35 year old arrestees.

Although these results show whether PSA provision leads to a harsher or more lenient decision (and whether it increases or decreases the proportions of negative outcomes), they are not informative about whether it helps judges make better decisions. In the current context, a primary goal of the judge is to make lenient decisions in low-risk cases and less lenient decisions in high-risk cases. If the PSA is helpful, therefore, its provision should encourage the judge to impose small or no bail on safe cases and impose a greater amount of bail on risky cases (we formally define 'safe' and 'risky' cases below). This demands the study of an important causal heterogeneity by distinguishing among cases with different risk levels. In addition, we may also be interested in knowing how PSA provision affects the gender and racial fairness of judges' decisions. Thus, the goal of the remainder of the paper is to develop statistical methods that directly address these and other questions.

3 The proposed evaluation methodology

In this section, we describe the proposed methodology for experimentally evaluating the impacts of algorithmic recommendations on human decision-making. Although we refer to our specific application throughout, the proposed methodology can be applied or extended to other settings, in

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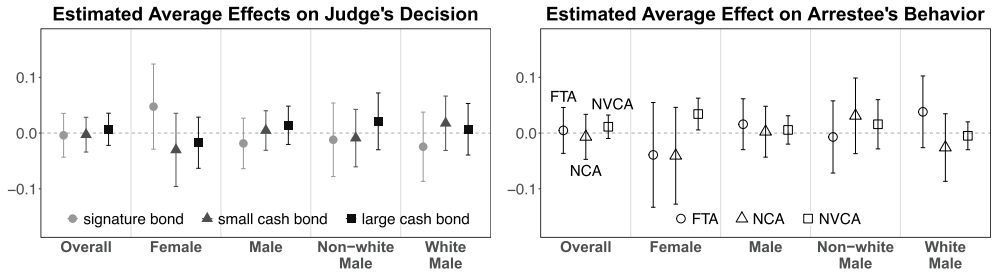

Figure 2. Estimated average causal effects of PSA provision on the judge's decisions and outcome variables. The results are based on the difference-in-means estimator. The vertical bars represent the 95% confidence intervals. In the left plot, we report the estimated effects of PSA provision on the judge's decision to charge a signature bond (solid circles), a small cash bail ($1,000 dollars or less; solid triangles), and a large cash bail (greater than $1,000; solid squares). In the right plot, we report the estimated effects of PSA provision on the three different outcome variables: FTA (open circles), NCA (open triangles), and NVCA (open squares). PSA provision appears to have little overall effect on the judge's decision and arrestee's behaviour, on average, though it may slightly increase NVCA among female arrestees.

which humans make decisions using algorithmic recommendations as an input. We will begin by considering a binary decision and then extend our methodology to an ordinal decision in Section 3.4.

3.1 The setup

Let Z_i be a binary treatment variable indicating whether the PSA is presented to the judge of case $i = 1, 2, \ldots, n$. We use D_i to denote the binary detention decision made by the judge to either detain ($D_i = 1$) or release ($D_i = 0$) the arrestee prior to the trial. In addition, let Y_i represent the binary outcome: we code all our outcomes—NCA, NVCA, and FTA—as binary variables. For example, $Y_i = 1$ ($Y_i = 0$) implies that the arrestee of case i commits (does not commit) an NCA. Finally, we use $\mathbf{X}_i$ to denote a vector of observed pre-treatment covariates for case i. They include age, gender, race, and prior criminal history.

We adopt the potential outcomes framework of causal inference and assume the stable unit treatment value assumption (SUTVA) (Rubin, 1990). In particular, we assume no interference among cases, implying that the treatment assignment for one case does not influence the judge's decision and outcome variable in another case. This assumption is reasonable in our analysis because we focus only on first arrests and do not analyse cases with subsequent arrests. Online Supplementary Material, Appendix S3 provides the empirical evidence in support of this assumption.

Let $D_i(z)$ be the potential value of the pretrial detention decision if case i is assigned to the treatment condition $z \in \{0, 1\}$. Furthermore, $Y_i(z, d)$ represents the potential outcome under the scenario, in which case i is assigned to the treatment condition z and the judge makes the decision $d \in \{0, 1\}$. Then, the observed decision is given by $D_i = D(Z_i)$, whereas the observed outcome is denoted by $Y_i = Y_i(Z_i, D_i(Z_i))$.

Throughout this paper, we maintain the following three assumptions, all of which we believe are reasonable in our application. First, because the treatment assignment is essentially randomized, the following independence assumption is automatically satisfied.

Assumption 1 (Randomization of the Treatment Assignment).

$$\{D_i(z), Y_i(z, d), \mathbf{X}_i\} \perp\!\!\!\perp Z_i \quad \text{for } z \in \{0, 1\} \text{ and all } d.$$

Second, we assume that the provision of the PSA influences the outcome only through the judge's decision. Because an arrestee would not care and, perhaps, would not even know whether the judge is presented with the PSA at their first appearance, it is reasonable to assume that their behaviour, be it NCA, NVCA, or FTA, is not affected directly by the treatment assignment.

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Assumption 2 (Exclusion Restriction).

$$Y_i(z, d) = Y_i(z', d) \quad \text{for } z, z' \in \{0, 1\} \text{ and all } i, d.$$

Under Assumption 2, we can simplify our notation by writing $Y_i(z, d)$ as $Y_i(d)$. A potential violation of this assumption is that the PSA may directly influence the judge's decision about release conditions, which can in turn affect the outcome. The extension of the proposed methodology to multi-dimensional decisions, including the bail amount and monitoring conditions, is left for future research.

Finally, we assume that the judge's decision monotonically affects the outcome. Thus, for NCA (NVCA), the assumption implies that each arrestee is no less likely to commit a new (violent) crime if released. If FTA is the outcome of interest, this assumption implies that an arrestee is no more likely to appear in court if released. The assumption is reasonable because being held in custody of a court makes it difficult to engage in NCA, NVCA, and FTA.

Assumption 3 (Monotonicity).

$$Y_i(1) \leq Y_i(0) \quad \text{for all } i.$$

3.2 Causal quantities of interest

We define causal quantities of interest using principal strata that are determined by the joint values of potential outcomes, i.e., $(Y_i(1), Y_i(0)) = (y_1, y_0)$, where $y_1, y_0 \in \{0, 1\}$ (Frangakis & Rubin, 2002). Since Assumption 3 eliminates one principal stratum, $(Y_i(1), Y_i(0)) = (1, 0)$, there are three remaining principal strata. The stratum $(Y_i(1), Y_i(0)) = (0, 1)$ consists of those who would engage in NCA (NVCA or FTA) only if they are released. We call members of this stratum as 'preventable cases' because keeping those arrestees in custody would prevent the negative outcome (NCA, NVCA, or FTA). The stratum $(Y_i(1), Y_i(0)) = (1, 1)$ is called 'risky cases' and corresponds to those who always engage in NCA (NVCA or FTA) regardless of the judge's decision. In contrast, the stratum $(Y_i(1), Y_i(0)) = (0, 0)$ represents 'safe cases' in which the arrestees would never engage in NCA (NVCA or FTA) regardless of the detention decision.

We are interested in examining how PSA provision influences the judge's detention decisions across different types of cases. We define the following three average principal causal effects (APCE),

$$\text{APCEp} = \mathbb{E}\{D_i(1) - D_i(0) \mid Y_i(1) = 0, Y_i(0) = 1\}, \tag{1}$$

$$\text{APCEr} = \mathbb{E}\{D_i(1) - D_i(0) \mid Y_i(1) = 1, Y_i(0) = 1\}, \tag{2}$$

$$\text{APCEs} = \mathbb{E}\{D_i(1) - D_i(0) \mid Y_i(1) = 0, Y_i(0) = 0\}. \tag{3}$$

If the PSA is helpful, its provisions should make the judge more likely to detain the arrestees of preventable cases. That is, the principal causal effect on the detention decision for the preventable cases (APCEp) should be positive. In addition, the PSA should encourage the judge to release the arrestees of the safe cases, implying that the principal causal effect for the safe cases (APCEs) should be negative. The desirable direction of the principal causal effect for risky cases (APCEr) depends on various factors including the societal costs of holding the arrestees of this category in custody.

3.3 Nonparametric identification

We consider the nonparametric identification of the principal causal effects defined above. The following theorem shows that under the aforementioned assumptions, these effects can be identified up to the marginal distributions of $Y_i(d)$ for $d = 0, 1$.

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Theorem 1 (Identification). Under Assumptions 1, 2, and 3,

$$APCEp = \frac{\Pr(Y_i = 1 \mid Z_i = 0) - \Pr(Y_i = 1 \mid Z_i = 1)}{\Pr\{Y_i(0) = 1\} - \Pr\{Y_i(1) = 1\}},$$
$$APCEr = \frac{\Pr(D_i = 1,\ Y_i = 1 \mid Z_i = 1) - \Pr(D_i = 1,\ Y_i = 1 \mid Z_i = 0)}{\Pr\{Y_i(1) = 1\}},$$
$$APCEs = \frac{\Pr(D_i = 0,\ Y_i = 0 \mid Z_i = 0) - \Pr(D_i = 0,\ Y_i = 0 \mid Z_i = 1)}{1 - \Pr\{Y_i(0) = 1\}}.$$

Proof is given in Online Supplementary Material, Appendix S4.2. Because $\Pr\{Y_i(d)\}$ is not identifiable without additional assumptions, we cannot estimate the causal effects based on Theorem 1. The denominators of the expressions on the right-hand side of Theorem 1, however, are positive under Assumption 3. As a result, the signs of the causal effects are identified from Theorem 1, which allows us to draw qualitative conclusions. In addition, the theorem implies that the sign of APCEp is the opposite of the sign of the average causal effect on the outcome. This is intuitive because if the provision of the PSA increases the probability of NCA (NVCA or FTA), then the judge must have released more arrestees for preventable cases.

Furthermore, we can obtain the nonparametric bounds on these causal quantities by bounding $\Pr\{Y_i(d) = y\}$ that appears in the denominators. From Assumption 1 and the law of total probability

$$\begin{aligned}\Pr\{Y_i(d) = 1\} &= \Pr\{Y_i(d) = 1 \mid Z_i = z\}\\ &= \Pr(Y_i = 1 \mid D_i = d, Z_i = z)\Pr(D_i = d \mid Z_i = z)\\ &\quad + \Pr\{Y_i(d) = 1 \mid D_i = 1 - d, Z_i = z\}\Pr(D_i = 1 - d \mid Z_i = z)\end{aligned}$$

for z, $d = 0$, 1. Under Assumption 3, the bounds on the unidentifiable terms are $\Pr\{Y_i = 1 \mid D_i = 1, Z_i = z\} \le \Pr\{Y_i(0) = 1 \mid D_i = 1, Z_i = z\} \le 1$ and $0 \le \Pr\{Y_i(1) = 1 \mid D_i = 0, Z_i = z\} \le \Pr\{Y_i = 1 \mid D_i = 0, Z_i = z\}$. This yields the following bounds on $\Pr\{Y_i(d) = 1\}$,

$$\max_z \Pr(Y_i = 1, D_i = 1 \mid Z_i = z) \le \Pr\{Y_i(1) = 1\} \le \min_z \Pr(Y_i = 1 \mid Z_i = z),$$
$$\max_z \Pr(Y_i = 1 \mid Z_i = z) \le \Pr\{Y_i(0) = 1\} \le 1 - \max_z \Pr(Y_i = 0, D_i = 0 \mid Z_i = z).$$

For point identification, we consider the following unconfoundedness assumption, which states that conditional on a set of observed pre-treatment covariates $\mathbf{X}_i$ and PSA provision, the judge's decision is independent of the potential outcomes.

Assumption 4 (Unconfoundedness).

$$Y_i(d) \perp\!\!\!\perp D_i \mid \mathbf{X}_i = \mathbf{x}, Z_i = z,$$

where we also assume $0 < \Pr(D_i = d \mid \mathbf{X}_i = \mathbf{x}, Z_i = z) < 1$ for $z \in \{0, 1\}$, and all $\mathbf{x} \in \mathcal{X}$ and d.

Assumption 4 holds if $\mathbf{X}_i$ contains all the information the judge has access to when making the detention decision under each treatment condition. As noted in Section 2.2, however, the judge may receive and use additional information regarding whether the arrestee has a job or a family in the jurisdiction, or perhaps regarding the length of time the arrestee has lived in the jurisdiction. If these factors have an impact on both the judge's decisions and arrestee's behaviours, then the

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assumption is unlikely to be satisfied. Later, we address this issue by developing a sensitivity analysis for the potential violation of Assumption 4 (see Section 3.5).

To derive the identification result, consider the following principal scores (Ding & Lu, 2017), which represent in our application the population proportion (conditional on $\mathbf{X}_i$) of preventable, risky, and safe cases, respectively,

$$
\begin{aligned}
e_P(\mathbf{x}) &= \Pr\{Y_i(1) = 0,\ Y_i(0) = 1 \mid \mathbf{X}_i = \mathbf{x}\},\\
e_R(\mathbf{x}) &= \Pr\{Y_i(1) = 1,\ Y_i(0) = 1 \mid \mathbf{X}_i = \mathbf{x}\},\\
e_S(\mathbf{x}) &= \Pr\{Y_i(1) = 0,\ Y_i(0) = 0 \mid \mathbf{X}_i = \mathbf{x}\}.
\end{aligned}
$$

Under Assumptions 2, 3, and 4, we can identify the principal scores as,

$$
\begin{aligned}
e_P(\mathbf{x}) &= \Pr\{Y_i = 1 \mid D_i = 0, \mathbf{X}_i = \mathbf{x}\} - \Pr\{Y_i = 1 \mid D_i = 1, \mathbf{X}_i = \mathbf{x}\},\\
e_R(\mathbf{x}) &= \Pr\{Y_i = 1 \mid D_i = 1, \mathbf{X}_i = \mathbf{x}\},\\
e_S(\mathbf{x}) &= \Pr\{Y_i = 0 \mid D_i = 0, \mathbf{X}_i = \mathbf{x}\}.
\end{aligned}
$$

The next theorem shows that we can identify the APCE as the difference in the weighted average of judge's decisions between the treatment and control groups.

Theorem 2 (Identification under Unconfoundedness). Under Assumptions 1, 2, 3, and 4, APCEp, APCEr and APCEs are identified as,

$$
\begin{aligned}
APCEp &= \mathbb{E}\{w_P(\mathbf{X}_i)D_i \mid Z_i = 1\} - \mathbb{E}\{w_P(\mathbf{X}_i)D_i \mid Z_i = 0\},\\
APCEr &= \mathbb{E}\{w_R(\mathbf{X}_i)D_i \mid Z_i = 1\} - \mathbb{E}\{w_R(\mathbf{X}_i)D_i \mid Z_i = 0\},\\
APCEs &= \mathbb{E}\{w_S(\mathbf{X}_i)D_i \mid Z_i = 1\} - \mathbb{E}\{w_S(\mathbf{X}_i)D_i \mid Z_i = 0\},
\end{aligned}
$$

where

$$
w_P(\mathbf{x}) = \frac{e_P(\mathbf{x})}{\mathbb{E}\{e_P(\mathbf{X}_i)\}}, \quad w_R(\mathbf{x}) = \frac{e_R(\mathbf{x})}{\mathbb{E}\{e_R(\mathbf{X}_i)\}}, \quad w_S(\mathbf{x}) = \frac{e_S(\mathbf{x})}{\mathbb{E}\{e_S(\mathbf{X}_i)\}}.
$$

Proof is given in Online Supplementary Material, Appendix S4.2. Although Ding and Lu (2017) also identify principal causal effects using principal scores, they consider principal strata based on an intermediate variable. In contrast, we are interested in the causal effects on the decision within each principal stratum defined by the potential values of the outcome.

In some situations, we might consider the following strong monotonicity assumption instead of Assumption 3.

Assumption 5 (Strong Monotonicity).

$$
Y_i(1) = 0 \quad \text{for all } i.
$$

The assumption implies that the detention decision prevents NCA, NVCA, or FTA. The assumption is plausible for FTA, but may not hold for NCA/NVCA in some cases. In our data, for example, we find some NCA and NVCA among the incarcerated arrestees.

Under Assumption 5, the risky cases do not exist and hence the APCEr is not defined. This leads to the following identification result.

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Theorem 3 (Identification under Strong Monotonicity). Under Assumptions 1, 2, and 5,

$$APCEp = \frac{\Pr(D_i = 0,\ Y_i = 1 \mid Z_i = 0) - \Pr(D_i = 0,\ Y_i = 1 \mid Z_i = 1)}{\Pr\{Y_i(0) = 1\}},$$
$$APCEs = \frac{\Pr(D_i = 0,\ Y_i = 0 \mid Z_i = 0) - \Pr(D_i = 0,\ Y_i = 0 \mid Z_i = 1)}{\Pr\{Y_i(0) = 0\}}.$$

Proof is given in Online Supplementary Material, Appendix S4.4. As in Theorem 1, the APCEp and APCEs depend on the distribution of $Y_i(0)$, which is not identifiable. However, as before, the sign of each effect is identifiable.

For point identification, we invoke the unconfoundedness assumption. Note that under the strong monotonicity assumption, Assumption 4 is equivalent to a weaker conditional independence relation concerning only one of the two potential outcomes,

$$Y_i(0) \perp\!\!\!\perp D_i \mid \mathbf{X}_i, Z_i = z$$

for $z = 0, 1$. We now present the identification result.

Theorem 4 (Identification under Unconfoundedness and Strong Monotonicity). Under Assumptions 1, 2, 4 and 5,

$$APCEp = \mathbb{E}\{w_P(\mathbf{X}_i)D_i \mid Z_i = 1\} - \mathbb{E}\{w_P(\mathbf{X}_i)D_i \mid Z_i = 0\},$$
$$APCEs = \mathbb{E}\{w_S(\mathbf{X}_i)D_i \mid Z_i = 1\} - \mathbb{E}\{w_S(\mathbf{X}_i)D_i \mid Z_i = 0\},$$

where

$$w_P(\mathbf{x}) = \frac{e_P(\mathbf{x})}{\mathbb{E}\{e_P(\mathbf{X}_i)\}}, \quad w_S(\mathbf{x}) = \frac{e_S(\mathbf{x})}{\mathbb{E}\{e_S(\mathbf{X}_i)\}}.$$

Proof is straightforward and hence omitted. While the identification formulas are identical to those in Theorem 2, under Assumption 5, we can simply compute the principal score as $e_S(\mathbf{x}) = \Pr(Y_i = 0 \mid D_i = 0, \mathbf{X}_i = \mathbf{x})$ and set $e_P(\mathbf{x}) = 1 - e_S(\mathbf{x})$.

3.4 Ordinal decision

We generalize the above identification results to an ordinal decision. In our application, this extension is important as the judge's release decision often is based on different amounts of cash bail or varying levels of supervision of an arrestee. We first generalize the monotonicity assumption (Assumption 3) by requiring that a decision with a greater amount of bail is no less likely to make an arrestee engage in NCA (NVCA or FTA). The assumption may be reasonable, for example, because a greater amount of bail is expected to imply a greater probability of being held in custody. The assumption could be violated if arrestees experience financial strain in an effort to post bail, causing them to commit NCA (NVCA or FTA).

Formally, let D_i be an ordinal decision variable where $D_i = 0$ is the least amount of bail, and $D_i = 1, \ldots, k$ represents a bail of increasing amount, i.e., $D_i = k$ is the largest bail amount. Then, the monotonicity assumption for an ordinal decision is given by,

Assumption 6 (Monotonicity with Ordinal Decision)

$$Y_i(d_1) \leq Y_i(d_2) \quad \text{for } d_1 \geq d_2.$$

To generalize the principal strata introduced in the binary decision case, we define the decision with the least amount of bail that prevents an arrestee from committing NCA (NVCA or FTA) as

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follows,

$$R_i = \begin{cases} \min\{d : Y_i(d) = 0\} & \text{if } Y_i(k) = 0, \\ k+1 & \text{if } Y_i(k) = 1. \end{cases}$$

We may view R_i as an ordinal measure of risk with a greater value indicating a higher degree of risk. When D_i is binary, R_i takes one of the three values, {0, 1, 2}, representing safe, preventable, and risky cases, respectively. Thus, R_i generalizes the principal strata to the ordinal case under the monotonicity assumption.

Now, we define the principal causal effects in the ordinal decision case. Specifically, for $r = 1, \ldots, k$ (excluding the cases with $r = 0$ and $r = k + 1$), we define the average principal causal effect of the PSA on the judge's decisions as a function of this ordinal risk measure,

$$\text{APCEp}(r) = \Pr\{D_i(1) \geq r \mid R_i = r\} - \Pr\{D_i(0) \geq r \mid R_i = r\}. \tag{4}$$

Since the arrestees with $R_i = r$ would not commit NCA (NVCA or FTA) under the decision with $D_i \geq r$, APCEp(r) represents a reduction in the proportion of NCA (NVCA or FTA) that is attributable to PSA provision among the cases with $R_i = r$. Thus, the expected proportion of NCA (NVCA or FTA) that would be reduced by the PSA is given by,

$$\sum_{r=1}^{k} \text{APCEp}(r) \cdot \Pr(R_i = r).$$

This quantity equals the overall Intention-to-Treat (ITT) effect of PSA provision on NCA (NVCA or FTA).

Furthermore, the arrestees with $R_i = 0$ would never commit a new crime regardless of the judge's decisions. We may, therefore, be interested in estimating the increase in the proportion of the most lenient decision for these safest cases. This generalizes the APCEs to the ordinal decision case,

$$\text{APCEs} = \Pr\{D_i(1) = 0 \mid R_i = 0\} - \Pr\{D_i(0) = 0 \mid R_i = 0\}.$$

For the cases with $R_i = k + 1$ that would always result in a new criminal activity, a desirable decision may depend on a number of factors. Note that if we assume the strong monotonicity, i.e., $Y_i(k) = 0$ for all i, then such cases do not exist.

Like the APCEs, the APCEp(r) can be expressed as a function of the average principal causal effect (APCE) for each decision $d = 0, 1, 2, \ldots, k$. This generalized APCE is given by,

$$\text{APCE}(d, r) = \Pr\{D_i(1) = d \mid R_i = r\} - \Pr\{D_i(0) = d \mid R_i = r\}. \tag{5}$$

In our empirical analysis, we estimate this causal quantity, which has the same identification conditions.

The identification of these principal causal effects requires the knowledge of the distribution of R_i. Fortunately, under the unconfoundedness and monotonicity assumptions (Assumptions 4 and 6), this distribution is identifiable conditional on $\mathbf{X}_i$,

$$\begin{aligned} e_r(\mathbf{x}) &= \Pr(R_i = r \mid \mathbf{X}_i = \mathbf{x}) \\ &= \Pr(R_i \geq r \mid \mathbf{X}_i = \mathbf{x}) - \Pr(R_i \geq r+1 \mid \mathbf{X}_i = \mathbf{x}) \\ &= \Pr\{Y_i(r-1) = 1 \mid \mathbf{X}_i = \mathbf{x}\} - \Pr\{Y_i(r) = 1 \mid \mathbf{X}_i = \mathbf{x}\} \\ &= \Pr\{Y_i = 1 \mid D_i = r-1, \mathbf{X}_i = \mathbf{x}\} - \Pr\{Y_i = 1 \mid D_i = r, \mathbf{X}_i = \mathbf{x}\}, \quad \text{for } r = 1, \ldots, k, \\ e_{k+1}(\mathbf{x}) &= \Pr\{Y_i(k) = 1 \mid \mathbf{X}_i = \mathbf{x}\} = \Pr\{Y_i = 1 \mid D_i = k, \mathbf{X}_i = \mathbf{x}\}, \\ e_0(\mathbf{x}) &= \Pr\{Y_i(0) = 0 \mid \mathbf{X}_i = \mathbf{x}\} = \Pr\{Y_i = 0 \mid D_i = 0, \mathbf{X}_i = \mathbf{x}\}. \end{aligned} \tag{6}$$

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Since $e_r(\mathbf{x})$ cannot be negative for each r, this yields a set of testable conditions for Assumptions 4 and 6. This statement is also true in the binary decision case.

Finally, we formally present the identification result for the ordinal decision case.

Theorem 5 (Identification with Ordinal Decision)

Under Assumptions 1, 2, 4 and 6, the APCE is identified by

$$\begin{aligned} APCEp(r) &= \mathbb{E}\{w_r(\mathbf{X}_i)\mathbf{1}(D_i \geq r) \mid Z_i = 1\} - \mathbb{E}\{w_r(\mathbf{X}_i)\mathbf{1}(D_i \geq r) \mid Z_i = 0\}, \\ APCEs &= \mathbb{E}\{w_0(\mathbf{X}_i)\mathbf{1}(D_i = 0) \mid Z_i = 1\} - \mathbb{E}\{w_0(\mathbf{X}_i)\mathbf{1}(D_i = 0) \mid Z_i = 0\}, \end{aligned}$$

where $w_r(\mathbf{x}) = e_r(\mathbf{x})/\mathbb{E}\{e_r(\mathbf{X}_i)\}$ and $\mathbf{1}()$ is the indicator function.

Proof is given in Online Supplementary Material, Appendix S4.5.

3.5 Sensitivity analysis

The unconfoundedness assumption, which enables the nonparametric identification of causal effects, may be violated when researchers do not observe some information that is used by the judge and is predictive of arrestees' behaviour. As noted in Section 2.2, the length of time the arrestee has lived in the community may represent an example of such unobserved confounders. It is important, therefore, to develop a sensitivity analysis for the potential violation of the unconfoundedness assumption (Assumption 4).

We propose a parametric sensitivity analysis (see Online Supplementary Material, Appendix S8 for nonparametric sensitivity analysis). We consider the following bivariate ordinal probit model for the observed judge's decision D and the latent risk measure R_i,

$$D_i^*(z) = \beta_Z z + \mathbf{X}_i^\top \beta_X + z\mathbf{X}_i^\top \beta_{ZX} + \epsilon_{i1}, \tag{7}$$

$$R_i^* = \mathbf{X}_i^\top \alpha_X + \epsilon_{i2}, \tag{8}$$

where

$$\begin{pmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right),$$

and

$$D_i(z) = \begin{cases} 0 & D^*(z) \leq \theta_{z1} \\ 1 & \theta_{z1} < D_i^*(z) \leq \theta_{z2} \\ \vdots & \vdots \\ k-1 & \theta_{z,k-1} < D_i^*(z) \leq \theta_{zk} \\ k & \theta_{zk} < D_i^*(z) \end{cases}, \quad R_i = \begin{cases} 0 & R_i^* \leq \delta_0 \\ 1 & \delta_0 < R_i^* \leq \delta_1 \\ \vdots & \vdots \\ k & \delta_{k-1} < R_i^* \leq \delta_k \\ k+1 & \delta_k < R_i^* \end{cases}.$$

The error terms $(\epsilon_{i1}, \epsilon_{i2})$ are assumed to follow a bivariate normal distribution. Under this model, ρ represents a sensitivity parameter since $\rho = 0$ implies Assumption 4. If the value of ρ is known, then the other coefficients, i.e., β_X, α_X and β_Z, can be estimated, which in turn enables the estimation of the APCE. In the literature, Frangakis et al. (2002), Barnard et al. (2003), and Forastiere et al. (2016) also model the distribution of principal strata using the ordinal probit model.

Because R_i is a latent variable, the estimation of this model is not straightforward. In our empirical application, we conduct a Bayesian analysis to estimate the causal effects (see e.g., Hirano et al., 2000; Jiang et al., 2016; Mattei et al., 2013; Schwartz et al., 2011, for other applications of Bayesian sensitivity analysis). Online Supplementary Material, Appendix S5 presents the details of the Bayesian estimation. We also perform a frequentist analysis, based on Theorem 2, that does

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not require an outcome model, assessing the robustness of the results to the outcome model (though we assume $\rho = 0$).

3.6 Fairness

Next, we discuss how the above causal effects relate to the fairness of the judge's decision. In particular, Imai and Jiang (2023) introduce the concept of 'principal fairness'. The basic idea is that within each principal stratum a fair decision should not depend on protected attributes (race, gender, etc.). Imai and Jiang (2023) provide a detailed discussion about how principal fairness is related to the existing definitions of fairness (see also Chouldechova & Roth, 2020; Corbett-Davies et al., 2017, and references therein). Although Coston et al. (2020) consider the potential outcomes framework, they only focus on one potential outcome $Y_i(0)$ rather than the joint potential outcomes $(Y_i(1), Y_i(0))$.

Formally, let $A_i \in \mathcal{A}$ be a protected attribute such as race and gender. We first consider a binary decision. We say that decisions are fair on average with respect to A_i if it does not depend on the attribute within each principal stratum, i.e.,

$$\Pr\{D_i = 1 \mid A_i,\ Y_i(1) = y_1,\ Y_i(0) = y_0\} = \Pr\{D_i = 1 \mid Y_i(1) = y_1,\ Y_i(0) = y_0\} \tag{9}$$

for all $y_1, y_0 \in \{0, 1\}$. We can generalize this definition to the ordinal case as,

$$\Pr(D_i \geq d \mid A_i,\ R_i = r) = \Pr(D_i \geq d \mid R_i = r)$$

for $1 \leq d \leq k$ and $0 \leq r \leq k + 1$.

The degree of fairness for principal stratum $R_i = r$ can be measured using the maximal deviation among the distributions for different groups,

$$\Delta_r(z) = \max_{a,a',d} \left| \Pr\{D_i(z) \geq d \mid A_i = a,\ R_i = r\} - \Pr\{D_i(z) \geq d \mid A_i = a',\ R_i = r\} \right| \tag{10}$$

for $z = 0, 1$. By estimating $\Delta_r(z)$, we can use the experimental data to examine whether or not the provision of the PSA improves the fairness of the judge's decisions. Specifically, if PSA provision improves the fairness of judge's decisions for the principal stratum r, we should have $\Delta_r(1) \leq \Delta_r(0)$.

3.7 Optimal decision rule

The discussion so far has focused on estimating the impacts of algorithmic recommendations on human decisions. We now show that the experimental data can also be used to derive an optimal decision rule given a certain objective. In addition, by comparing human decisions and algorithmic recommendations with optimal decision rules, we can evaluate their efficacy. In our application, one goal is to prevent as many NCAs (NVCAs or FTAs) as possible while avoiding unnecessarily harsh initial release decisions. To achieve this, we must carefully weigh the cost of negative outcomes and that of unnecessarily harsh decisions. Once these costs are specified as part of the utility function, one can empirically assess this tradeoff using the experimental data.

Formally, let δ be the judge's decision based on $\mathbf{X}_i$, which may include the PSA. We consider a deterministic decision rule, i.e., $\delta(\mathbf{x}) = d$ if $\mathbf{x} \in \mathcal{X}_d$ where $\mathcal{X}_d$ is a non-overlapping partition of the covariate space $\mathcal{X}$ with $\mathcal{X} = \bigcup_{r=0}^{k} \mathcal{X}_r$ and $\mathcal{X}_r \cap \mathcal{X}_{r'} = \emptyset$. We consider the utility function of the following form,

$$U_i(\delta) = \begin{cases} -c_0 & \delta(\mathbf{X}_i) < R_i \\ 1 & \delta(\mathbf{X}_i) = R_i \\ 1 - c_1 & \delta(\mathbf{X}_i) > R_i \end{cases},$$

where c_0 and c_1 represent the cost of an NCA (NVCA or FTA) and that of an unnecessarily harsh decision, respectively. Under this setting, preventing an NCA (NVCA or FTA) with the most lenient decision ($\delta(\mathbf{X}_i) = R_i$) yields the utility of one, while we incur the cost c_1 for an unnecessarily harsh decision ($\delta(\mathbf{X}_i) > R_i$), leading to the net utility of $1 - c_1$.

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The relative magnitude of these two cost parameters, c_0 and c_1, may depend on the consideration of various factors, including the potential harm to the public and arrestees caused by the negative outcomes and unnecessarily harsh decisions, respectively. When $c_0 = c_1 = 0$, for example, $U_i(\delta)$ reduces to $\mathbf{1}\{\delta(\mathbf{X}_i) \geq R_i\}$, which is non-zero only if the decision is sufficiently harsh so that it prevents the negative outcome. The optimal decision under this utility is the most stringent decision, i.e., $\delta(\mathbf{X}_i) = k$, for all cases. If $c_0 = 2$ and $c_1 = 1$, the resulting utility function implies that the cost of NCA (NVCA or FTA) is twice as large as that of an unnecessarily harsh decision.

We derive the optimal decision rule δ^* that maximizes the expected utility,

$$\delta^* = argmax_\delta \mathbb{E}\{U_i(\delta)\}.$$

For $r = 0, \ldots, k+1$ and $d = 0, \ldots, k$, we can write,

$$\mathbb{E}[\mathbf{1}\{\delta(\mathbf{X}_i) = d, R_i = r\}] = \mathbb{E}\{\mathbf{1}(\mathbf{X}_i \in \mathcal{X}_d, R_i = r)\} = \mathbb{E}\big\{\mathbf{1}(\mathbf{X}_i \in \mathcal{X}_d) \cdot e_r(\mathbf{X}_i)\big\}.$$

Thus, we can express the expected utility as,

$$\begin{aligned}
&\mathbb{E}\{U_i(\delta)\} \\
&= \sum_{r=0}^{k+1} \Bigg(\sum_{d>r} (1 - c_1) \mathbb{E}[\mathbf{1}\{\delta(\mathbf{X}_i) = d, R_i = r\}] + \sum_{d=r} \mathbb{E}[\mathbf{1}\{\delta(\mathbf{X}_i) = d, R_i = r\}] \\
&\quad - \sum_{d<r} c_0 \mathbb{E}[\mathbf{1}\{\delta(\mathbf{X}_i) = d, R_i = r\}] \Bigg) \qquad (11) \\
&= \sum_{r=0}^{k+1} \Bigg[\sum_{d \geq r} \mathbb{E}\big\{\mathbf{1}(\mathbf{X}_i \in \mathcal{X}_d) \cdot e_r(\mathbf{X}_i)\big\} - c_0 \sum_{d<r} \mathbb{E}\big\{\mathbf{1}(\mathbf{X}_i \in \mathcal{X}_d) \cdot e_r(\mathbf{X}_i)\big\} \\
&\quad - c_1 \sum_{d>r} \mathbb{E}\big\{\mathbf{1}(\mathbf{X}_i \in \mathcal{X}_d) \cdot e_r(\mathbf{X}_i)\big\} \Bigg] \\
&= \sum_{d=0}^{k} \mathbb{E}\Bigg[\mathbf{1}(\mathbf{X}_i \in \mathcal{X}_d) \Bigg\{ \sum_{r \leq d} e_r(\mathbf{X}_i) - c_0 \cdot \sum_{r>d} e_r(\mathbf{X}_i) - c_1 \cdot \sum_{r<d} e_r(\mathbf{X}_i) \Bigg\} \Bigg]. \qquad (12)
\end{aligned}$$

This yields the following optimal decision,

$$\delta^*(\mathbf{x}) = \underset{d \in \{0,\ldots,k\}}{\operatorname{argmax}}\, g_d(\mathbf{x}) \quad \text{where } g_d(\mathbf{x}) = \sum_{r \leq d} e_r(\mathbf{x}) - c_0 \cdot \sum_{r>d} e_r(\mathbf{x}) - c_1 \cdot \sum_{r<d} e_r(\mathbf{x}). \qquad (13)$$

We can, therefore, use the experimental estimate of $e_r(\mathbf{x})$ to learn about the optimal decision.

Policy makers could derive the optimal decision rule by using the above result and then adopt this rule as the recommendation for judges. However, this may not be useful if the judge decides to follow the algorithmic recommendation selectively for some cases or ignore it altogether. Instead, we may wish to construct PSA scores that maximize the optimality of the judge's decision. Unfortunately, the derivation of such an optimal PSA score is challenging since the PSA scores were not directly randomized in our experiment. We tackle this problem in a separate paper (Ben-Michael et al., 2021). In Online Supplementary Material, Appendix S6, we also consider the optimal provision of the PSA given the same goal considered above (i.e., prevent as many NCAs (NVCAs or FTAs) as possible with a minimal amount of bail).

4 Empirical analysis

In this section, we apply the proposed methodology to the data from the field RCT described in Section 2.

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4.1 Preliminaries

As explained in Section 2.3, we use the ordinal decision variable with three categories—the signature bond ($D_i = 0$), the bail amount of \$1,000 or less ($D_i = 1$), and the bail amount of greater than \$1,000 ($D_i = 2$). Given this ordinal decision, we label the principal strata as safe ($R_i = 0$), easily preventable ($R_i = 1$), preventable ($R_i = 2$), and risky cases ($R_i = 3$).

We fit the Bayesian model defined in Equations (7) and (8) with a diffuse prior distribution as specified in Online Supplementary Material, Appendix S5, separately for each of three binary outcome variables—FTA, NCA, and NVCA. The model incorporates the following pre-treatment covariates: gender (male or female), race (white or non-white), the interaction between gender and race, age, and several indicator variables regarding the current and past charges. It also includes a binary variable for the presence of pending charge (felony, misdemeanor, or both) at the time of offense, four binary variables for current charges (non-violent misdemeanor, violent misdemeanor, non-violent felony, and violent felony), a four-level ordinal variable for the DMF recommendation, three variables for prior conviction (binary variables for misdemeanor and felony as well as a four-level ordinal variable for violent conviction), a binary variable for prior sentence to incarceration, and two variables for prior FTAs (a three-level ordinal variable for FTAs from past two years, and a binary variable for FTAs from over two years ago).

We use the Gibbs sampling and run five Markov chains of 100,000 iterations each with random starting values independently drawn from the prior distribution. Based on the Gelman-Rubin statistic for convergence diagnostics, we retain the second half of each chain and combine them to be used for our analysis. Online Supplementary Material, Appendix S5 presents the computational details including the Gibbs sampling algorithm we use.

We begin by computing the estimated population proportion of each principal stratum based on Equation (6). Figure 3 presents the results. We find that for FTA, the overall proportion of safe cases (blue) is estimated to be 67%, whereas those of easily preventable (black), preventable (red), and risky (brown) cases are 6%, 7%, and 20%, respectively. A similar pattern is observed for FTA and NCA across different racial and gender groups, while the estimated overall proportion of safe cases is even higher for NVCA, exceeding 90%.

4.2 Average principal causal effects

Figure 4 presents the estimated APCE of PSA provision on the three ordinal decision categories, separately for each of the three outcomes and each principal stratum (see Equation (5)). The overall and subgroup-specific results are given for each of the four principal strata—safe (blue), easily preventable (black), preventable (red), and risky (brown) cases. For a given principal stratum, we present the estimated APCE on each decision category—signature bond (circle), small cash bond (triangle), and large cash bond (square). The left column of each panel shows that PSA provision has little overall impact on the judge's decision across four principal strata for FTA and NCA. There is a suggestive, but inconclusive, evidence that PSA provision leads to an overall harsher decision for NVCA among easily preventable, preventable, and risky cases.

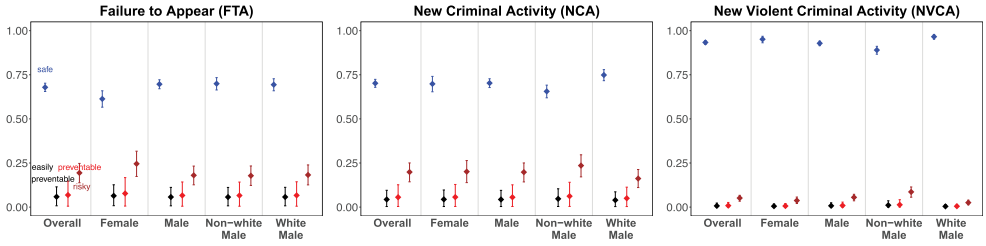

Figure 3. Estimated proportion of each principal stratum. Each plot represents the result using one of the three outcome variables (FTA, NCA, and NVCA), where the blue, black, red, and brown diamonds represent the estimates for safe, easily preventable, preventable, risky cases, respectively. The solid vertical lines represent the 95% Bayesian credible intervals. The results show that a vast majority of cases are safe across subgroups and across different outcomes. The proportion of safe cases is estimated to be especially high for NVCA.

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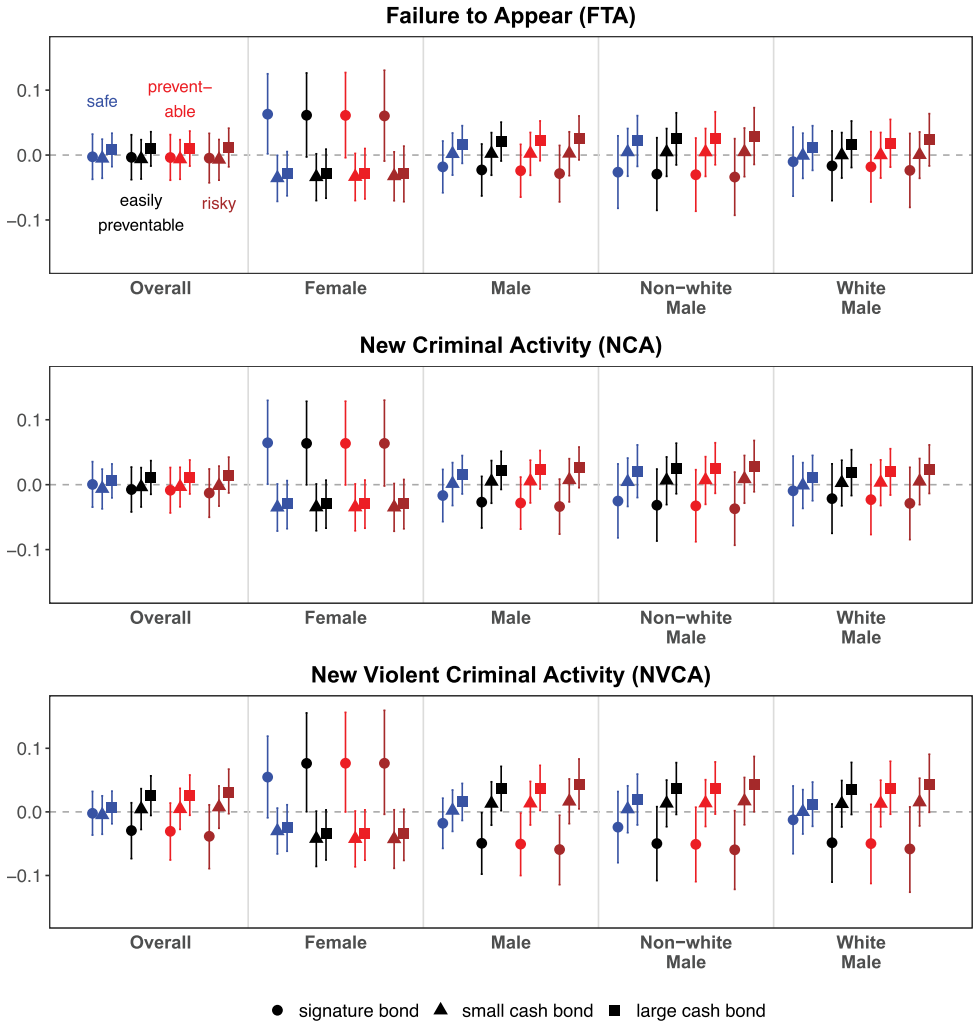

Figure 4. Estimated average principal causal effects (APCE) of PSA provision on the judge's decision. Each panel presents the overall and subgroup-specific results for a different outcome variable. Each column within a panel shows the estimated APCE of PSA provision for safe (blue), easily preventable (black), preventable (red), and risky (brown) cases. For each of these principal strata, we report the estimated APCE on the judge's decision to impose a signature bond (circles), a small cash bail amount of 1,000 dollars or less (triangles), and a large cash bail amount of greater than 1,000 (squares). The vertical line for each estimate represents the Bayesian 95% credible interval. The results show that PSA provision may make the judge's decision more lenient for female arrestees regardless of their risk levels. PSA provision may also encourage the judge to make harsher decisions for male arrestees with a greater risk level though the effect sizes are relatively small.

We also present the estimated APCE for different gender and racial groups in the remaining columns of each panel. We find potentially suggestive evidence that PSA provision may make it more likely for the judge to impose signature bonds (circles) on female arrestees instead of cash bonds (triangles and squares) across three outcomes. Interestingly, for all outcomes, this pattern appears to hold for any of the four principal strata, implying that PSA provision might not help the judge distinguish different risk levels of female arrestees. Our analysis also finds that for NVCA, PSA provision may lead to a harsher decision for easily preventable, preventable, and risky cases among male arrestees while it has little effect on the safe cases. This suggests that PSA provision may help distinguish different risk levels among male arrestees, resulting in improved decisions at least in terms of the original goal of the PSA. There is no discernible racial difference in these effects.

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In Online Supplementary Material, Appendix S2.2, we explore the estimated APCE for different age groups. We find that PSA provision may lead to a harsher decision for arrestees of the 29–35 years old group across three outcomes. This pattern appears to generally hold across all principal strata, though for NVCA, the effects are more pronounced for easily preventable, preventable, and risky cases. In addition, our analysis yields suggestive evidence that across all outcomes, PSA provision may make the judge's decision more lenient for the oldest (46 years old or above) group. This appears to be true across all three outcomes except that for NVCA the effect may exist only for safe cases. We reiterate, however, that these results are based on preliminary data and the effect sizes are relatively small.

We conduct two robustness analyses. Firstly, we perform a frequentist analysis that is based on Theorem 2 and does not assume an outcome model. The results are shown in Online Supplementary Material, Appendix S7 and are largely consistent with those shown here. As expected, the estimation uncertainty of the frequentist analysis, which makes less stringent assumptions than Bayesian analysis, is greater. Secondly, we conduct parametric sensitivity analyses using the methods described in Section 3.5 (see Online Supplementary Material, Appendix S9). We set the value of correlation parameter ρ to 0.05, 0.1, and 0.3 and examine how the estimated APCE changes. The results (see Online Supplementary Material, Figures S13–S15) are largely consistent across different values of ρ although the effects for females tend to exhibit a large degree of estimation uncertainty especially when the correlation is high and particularly for NVCA. This is not surprising. There are only a small number of female arrestees and only a handful of NVCA events corresponding to them.

4.3 Gender and racial fairness

We now examine the impacts of PSA provision on gender and racial fairness. Specifically, we evaluate the principal fairness of PSA provision as discussed in Section 3.6. We use gender (female vs. male) and race (white male vs. non-white male) separately as a protected attribute and analyse whether or not the provision of the PSA improves the fairness of the judge's decision in terms of the protected attribute. While the gender analysis is based on the entire sample, the racial analysis is based on the male sample only due to the limited sample size for females.

Figure 5 presents the results for gender (top panel) and racial (bottom panel) fairness across the principal strata and separately for each of the three outcomes. Each column within a given plot presents $\Delta_r(z)$ defined in equation (10), which represents the maximal subgroup difference in the judge's decision probability distribution within the same principal stratum $R_i = r$ under the provision of the PSA ($z = 1$) compared to no use of the PSA ($z = 0$). In this application, the maximal difference always occurs at $d = 1$, allowing us to interpret $\Delta_r(z)$ as the difference in the probability of imposing a cash bond ($D \geq 1$) rather than a signature bond. We also present the estimated difference caused by PSA provision in the two maximal subgroup differences, i.e., $\Delta_r(1) - \Delta_r(0)$. If this difference is estimated to be positive, then PSA provision reduces the fairness of judge's decisions by increasing the maximal subgroup difference.

We find that PSA provision might worsen the gender fairness of the judge's decisions. When the PSA is provided, the maximal gender difference in the judge's decision probability is on average greater than that when it is not provided. The effect is particularly large and statistically significant for NVCA and for preventable, easily preventable, and risky cases. This is consistent with our finding that especially for NVCA, PSA provision might make the judge's decision more lenient for female arrestees while it leads to a harsher decision for male arrestees among preventable, easily preventable, and risky cases. Thus, PSA provision appears to increase disparate decision-making across gender.

PSA provision, however, does not have a statistically significant impact on the racial fairness of the judges' decisions among male arrestees. For instance, in the principal stratum of safe cases, we find that PSA provision does not affect the maximal difference in the judge's decision probability (between non-white males and white males). This suggests that in terms of principal fairness, the PSA may not alter any existing racial difference in the judge's decisions.

4.4 Using optimal decision to evaluate the DMF recommendation

Finally, we evaluate the DMF recommendation by comparing it with the optimal decision under different values of the costs. For simplicity, we consider a binary decision: signature or cash bond.

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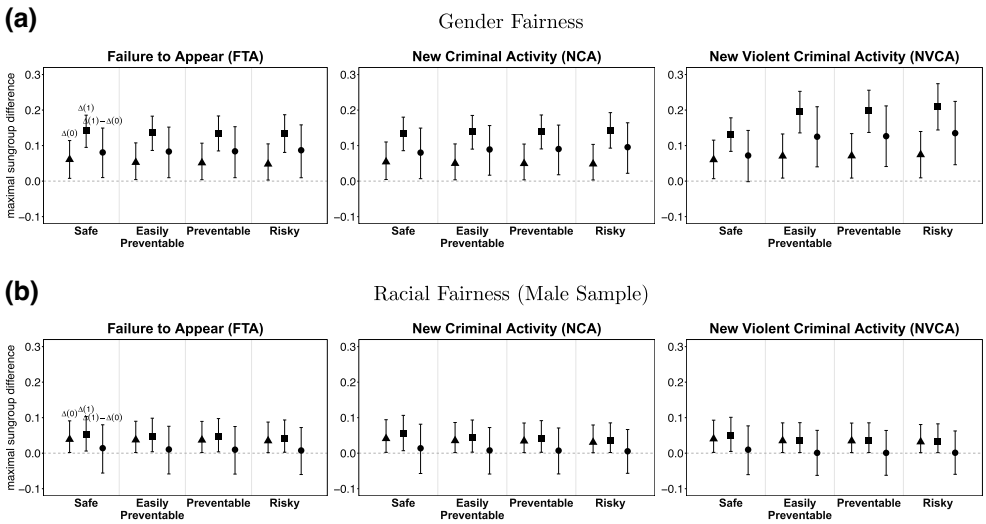

Figure 5. Gender and racial fairness of the judge's decisions. Within each plot, we show three estimates separately for each principal stratum—the maximal subgroup difference in the judge's decision probability of imposing a cash bond with PSA provision (squares; $\Delta(1)$) and without it (triangles; $\Delta(0)$) as well as the difference between them (circles; $\Delta(1) - \Delta(0)$). The vertical solid lines represent the 95% Bayesian credible intervals. A positive value of the difference would imply that the PSA reduces the fairness of the judge's decisions. For the gender analysis (top panel), even without the PSA, the judge seems to be more likely to impose a cash bond on male arrestees when compared to female arrestees with the same risk levels. PSA provision appears to increase this tendency. For the race analysis (bottom panel), PSA provision has little impact across all outcomes and risk levels. We reiterate, however, that these results are based on the preliminary data. (a) Gender fairness. (b) Racial fairness (male sample).

As discussed in Section 3.7, given a specific pair of cost parameters (c_0, c_1) and the experimental estimate of $e_r(\mathbf{x})$ for $r = 0, 1, 2$, we can compute the optimal decision for each case according to Equation (13). We then obtain the estimated proportion of cases, for which a cash bond is optimal. We repeat this process for a grid of different values for the cost of a negative outcome (c_0; FTA, NCA, and NVCA) and that of an unnecessarily harsh decision (c_1).

The top panel of Figure 6 presents the results for the cases whose DMF recommendation is a signature bond. In contrast, the bottom panel of the figure shows the results for the other cases (i.e., the DMF recommendation is a cash bond). In each plot, a darker grey region represents a greater proportion of cases, for which a cash bond is optimal. The results suggest that unless the cost of a negative outcome is much higher than the cost of an unnecessarily harsh decision, imposing a signature bond is the optimal decision for a vast majority of cases.

We also find that for all three outcomes, a cash bond is optimal for a greater proportion of cases when the DMF recommendation is indeed a cash bond. However, this difference is small, suggesting that the DMF recommendation is only mildly informative. Similar results are found even if we separately examine three PSA scores (see Figure S16 in Online Supplementary Material, Appendix S10).

4.5 Comparison between the judge's decisions and DMF recommendations

Lastly, we compare the judge's actual decision with the DMF recommendation in terms of the expected utility given in Equation (12). The top panel of Figure 7 represents the results for the treatment group (i.e., judge's decisions with the PSA), whereas the bottom panel represents those for the control group (i.e., judge's decisions without the PSA). A darker grey area indicates that the expected utility for the judge's decision is estimated to be greater than the DMF recommendation. Most of these estimates are statistically significant (see Online Supplementary Material, Figure S11 for more details). Therefore, unless the cost of a negative outcome is much greater than the cost of an unnecessarily harsh decision, the judge's decision (with or without the PSA scores) yields a greater expected utility than the DMF recommendation. This is especially true for NVCA.

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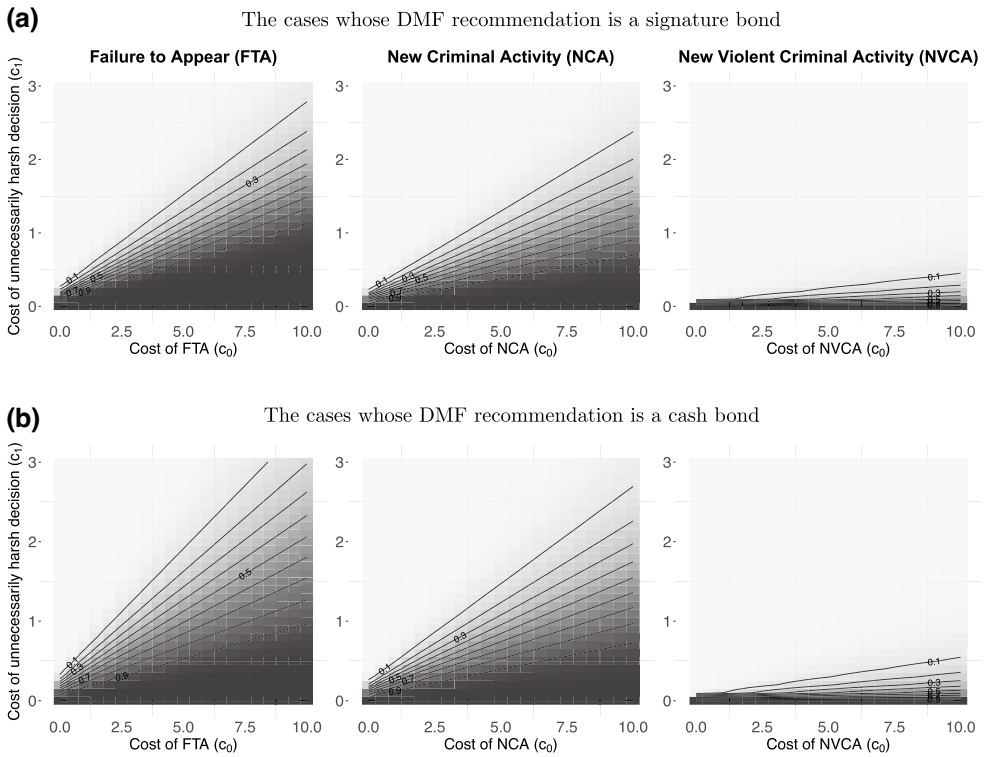

Figure 6. Estimated proportion of cases for which cash bond is optimal. Each column represents the results based on one of the three outcomes (FTA, NCA, and NVCA). The top (bottom) panel shows the results for the cases whose DMF recommendation is a signature (cash) bond. In each plot, the contour lines represent the estimated proportions of cases for which a cash bond is optimal, given the cost of an unnecessarily harsh decision (c_1; *y*-axis) and that of a negative outcome (c_0; *x*-axis). A dark grey area represents a greater proportion of such cases. The results show that regardless of DMF recommendation, a signature bond is optimal unless the cost of a negative outcome is much greater than the cost of an unnecessarily harsh decision. (a) The cases whose DMF recommendation is a signature bond. (b) The cases whose DMF recommendation is a cash bond.

Altogether, our analysis implies that the DMF recommendations may be unnecessarily harsher than the judge's decisions.

5 Concluding remarks

In today's data-rich society, many human decisions are guided by algorithmic recommendations. While some of these algorithmic-assisted human decisions may be trivial and routine (e.g., online shopping and movie suggestions), others that are much more consequential include judicial and medical decision-making. As algorithmic recommendation systems play increasingly important roles in our lives, we believe that a policy-relevant question is how such systems influence human decisions and how the biases of algorithmic recommendations interact with those of human decisions. These questions necessitate the empirical evaluation of the impacts of algorithmic recommendations on human decisions.

In this paper, we present a set of general statistical methods that can be used for the experimental evaluation of algorithm-assisted human decision-making. We applied these methods to the preliminary data from the first-ever randomized controlled trial for assessing the impacts of PSA provision on judges' pretrial decisions. There are several findings that emerge from our initial analysis. Firstly, we find that PSA provision has little overall impact on the judge's decisions. Secondly, we find potentially suggestive evidence PSA provision may encourage the judge to make more lenient decisions for female arrestees regardless of their risk levels while leading to more stringent decisions for males who are classified as risky. Thirdly, PSA provision appears to widen the existing

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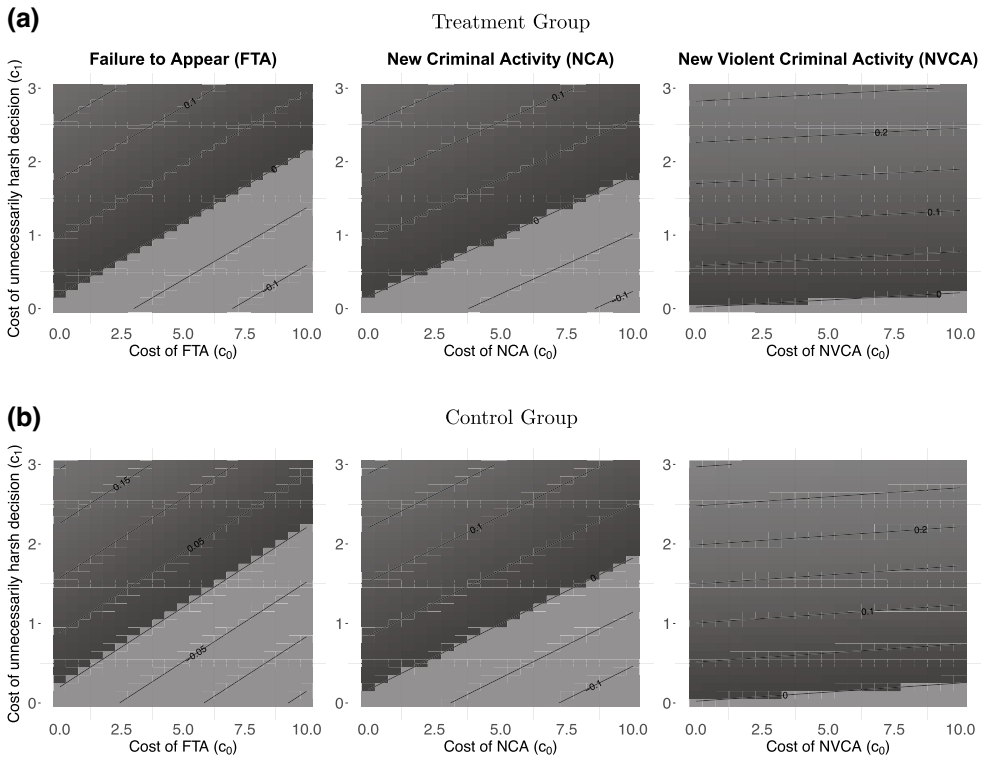

Figure 7. Estimated difference in the expected utility between judge's decisions and DMF recommendations for the treatment (top panel) and control (bottom panel) group. Each column represents the results based on one of the three outcomes, with a darker region indicating the values of the costs (the cost of a negative outcome and the cost of an unnecessarily harsh decision) for which the judge's decision yields a higher expected utility than the corresponding DMF recommendation. The results show that the judge's decision yields a higher expected utility than the DMF recommendation unless the cost of a negative outcome is much higher than that of an unnecessarily harsh decision. This pattern holds for all outcomes and is unchanged by the provision of the PSA. (a) Treatment group. (b) Control group.

gender difference of the judge's decisions against male arrestees, whereas it does not seem to alter decision-making across race among male arrestees. We caution, however, that these findings could be explained by other factors that are correlated with gender and race. Finally, we find that for a vast majority of cases, the optimal decision is to impose a signature bond rather than a cash bond unless the cost of a negative outcome is much higher than that of a decision that may result in unnecessary incarceration. This suggests that the PSA's recommendations may be harsher than necessary.

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Supplementary material

Supplementary material is available at *Journal of the Royal Statistical Society: Series A* online.

Conflict of interests: None declared.

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Discussion Paper Contribution

Proposer of the vote of thanks and contribution to the Discussion of 'Experimental evaluation of algorithm- assisted human decision-making: application to pretrial public safety assessment' by Imai et al.

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This article is a valuable contribution to the discussion of whether we should be using algorithms to make decisions in the criminal justice system. It provides two important causal results: when a judge was given recommendations from an algorithm this did not change the judge's decisions significantly, and that the criminal justice outcomes (failure to appear, new criminal activity, and new violent criminal activity) did not change with the algorithm's recommendations either. It is also important that we know this thanks to a randomised controlled trial (RCT), which is the gold standard for obtaining causal results, and it is a notoriously difficult study design to implement in this setting mostly because of logistical challenges. An RCT sets an example to researchers about how we could better evaluate policies in courts.

First, the RCT could be improved in several ways, and I do not think this is a mystery to them. The authors should include a larger number of judges and jurisdictions in their studies to avoid conclusions based on a single individual's decisions. During the discussion, the authors mentioned that they were continuing to perform studies, so perhaps they are addressing these issues in their current trials.

Second, I have some concerns about the use of principal strata in this application. The authors are interested not only in the criminal justice outcomes, but also in separating the arrestees depending on whether the arrestee case is risky. Their definition of the causal parameters was clear, as was their identification by making causal assumptions. I can understand why they would want to split the sample up into cases depending on risk based on the motivation of unnecessarily harsh decisions. However, I do not understand the policy relevance of these effects because we cannot tell in which stratum each individual lies.

I provide a directed acyclic graph in Figure 1 for clarity. The effect of PSA-DMF $\rightarrow$ D $\rightarrow$ Y answered the question: Did the algorithm help the judge achieve his or her goal of preventing arrestees from committing a new crime or failing to appear in court? The researchers' results were a null effect, although they found that PSA might increase new violent criminal activity among female arrestees, and possibly in 29–35 year old. This is an interesting result because it shows that the algorithm did not have a significant effect on the outcomes in the criminal justice system. In a later comment from James Greiner during the discussion, he said that the judge who had worked with them said (s)he thought the treatment would have an effect. I wonder, however, since the judge was aware that she was receiving the treatment, did this bias how she made decisions? This seems

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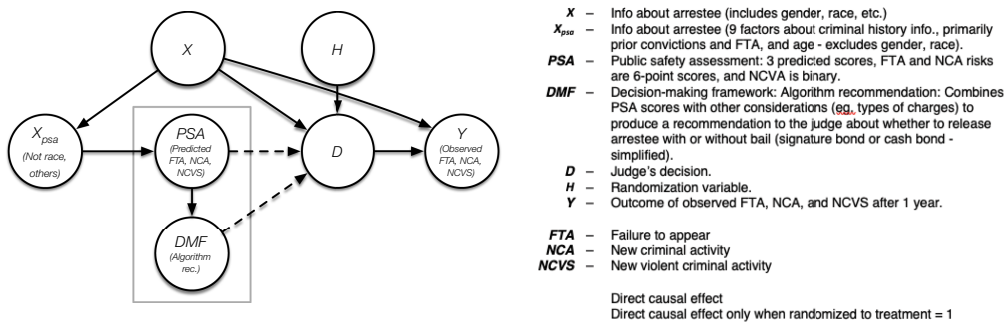

Figure 1. Directed acyclic graph, inspired by conversation with I. Ilya Shpitser (personal communication, February 7, 2022).

like an unavoidable problem, but perhaps a new experiment could be designed such that the decision-maker is unaware of whether they are receiving the treatment.

The effect of PSA-DMF → D answered the question: Did the PSA-DMF recommendation affect the judge's decisions? The researchers found a null effect for this as well: PSA provision has little effect on judge's decision. This result is even more interesting because it shows that the judges' decisions were not highly influenced by what the algorithm was telling them. Were the judges making the same predictions as the algorithm already? Note that they were observing the same information as the algorithm (namely the information about the arrestee, X_{psa}), and more. Or did they simply observe the results and think they knew better? It would be interesting to find out why the judge was not influenced by the algorithm.

Regarding fairness, one influential argument about why we should use algorithms is because they might help remove human biases. Humans can make biased decisions while being aware of it or not, and we could define an algorithm that does not do this. The argument is that whatever priors and psychological aspects affect us in terms of bias, would not affect the algorithm. But no matter how careful the algorithm is about this, the final decision can still be biased because race or gender can still affect the judge's decision D and the outcome Y directly. The hybrid results *include* human bias.

Finally, it was unclear why the researchers picked this definition of fairness. There is an extensive literature and discussion about which definition of fairness to use in computer science, machine learning, and statistics (e.g., see 21 fairness definitions and their politics by Arvind Narayanan), not to mention similar discussions in philosophy since ancient times. There should be a better argument for why this is the proper definition to choose, and furthermore it cannot be calculated for individuals, so this has the same problem as the principal strata causal effects from earlier. Furthermore, as in the argument above about principal strata, I do not see how we can get around the fact that the stratum it is unknowable at an individual level. If we cannot know it for an individual, then how can I use that notion to give a new arrestee a bail decision?

It would be interesting to run a similar RCT to isolate effects on outcomes of algorithm-only vs. human-only vs. hybrid. We should also compare whether the algorithm, human, or hybrid approach is more fair—after selecting a definition of fairness that is appropriate to this case (by talking to philosophers). Overall, this is an interesting and thought-provoking paper.

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Seconder of the vote of thanks and contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment' by Imai et al.

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RSS Discussion Meeting, 8 February 2022

To make a name for learning
when other roads are barred,
take something very easy
and make it very hard.
Piet Hein[1]

The use of algorithms to assist judicial decision-making is an important topic, but one which has been especially prone to serious statistical misunderstandings (Imrey & Dawid, 2015; Royal Statistical Society, 2018). To these interpretative misdemeanours, we must now add this monumental methodological misstep.

Notwithstanding the prevalence of arguments making essential use of 'principal strata', and notwithstanding that some of the proponents of these arguments have been honoured with the 2021 Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel, I continue to find all such approaches totally misguided (Dawid & Didelez, 2012). Has any inference that can only be expressed in terms of what happens to members of a specific principal stratum ever been checked empirically? How could it be, since we can never know which principal stratum an individual belongs to? How then can such an 'inference' possibly have any bearing on real-world decision-making? In the context of §3 of tonight's paper, does it make any sense to suppose that, prior to the hearing, a defendant already has a definite value (potential outcome) for whether or not he will appear at a subsequent hearing, for each of the various decisions the judge might make, all these potential outcomes having simultaneous existence? Only if we can accept this metaphysical conceit—which I cannot—can we even pose the sort of question addressed in this paper. And even then, the mathematical machinery requires that strong but logically unverifiable assumptions be made about the relationships between these potential outcomes, and then delivers (typically imprecise) 'inferences' relating to logically unknowable principal stratum membership (as in expressions (1)–(3) of the paper). What use can we possibly make of such 'knowledge'?

The paper does however raise two interesting substantive questions: In the light of the case information $X = x$ available to the judge:

(i) Given our understanding, from the data, of how judges and defendants behave, how should we choose Z? That is, should we present PSA to the judge before she makes her decision, or not?
(ii) How, ideally, should the judge use the information she has in coming to her decision D?

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Much as in §2.4 of the paper, each of these problems can be straightforwardly tackled using the well-developed machinery of statistical decision theory (Dawid, 2021; Raiffa, 1968). For (i), since Z is randomised we can estimate, from the data, the distribution of the outcome variable, Y or D, of interest, given a postulated choice of the value z of Z (and given the fixed background information $X = x$), and then choose $z = 0$ or 1 according as which of these two distributions we prefer. More formally, we can attach a utility value $u(y, x)$ to obtaining outcome y of Y for a case with characteristics x, and select z to maximise $\mathbb{E}\{u(Y, x) \mid Z = z, X = x\}$. Similarly for (ii): under the assumption of no confounding between D and Y, the optimal value d of D will maximise the (estimated) expected utility $\mathbb{E}\{u(Y, x) \mid D = d, X = x\}$ (where X now includes PSA, if supplied).

In contrast, the contorted novel approach of this paper, developed in §3, attaches importance, not merely to the eventual outcome Y, but to further unknowable aspects of the collection of potential responses. It evaluates the outcome, not just in terms of what it is, but also what (metaphysically) it might have been. For example, in the formal decision-theoretic approach to question (ii) in §3.7, the utility $U_i(\delta)$ is supposed to depend on the principal stratum indicator (riskiness) R_i, and thus on both the factual and the counterfactual outcomes. Even if I were to accept the existence of counterfactual 'might have beens,' I fail to see their relevance to decision problems. In the case at hand, 'riskiness' should be judged on the basis of the known characteristics x of the case, and how these impact probabilistic uncertainty about the outcome Y, for each available decision—not in terms of some unknowable and meaningless potential outcomes.

In an attempt to salvage something from the shipwreck, one might consider an additional variable V, which is not observed before a decision has to be taken (and thus is not part of X), but whose value v is considered relevant to the utility of the outcome, which is thus now of the form $u(y, v, x)$. Then in problem (ii), for example, one would have to choose d to maximise $\mathbb{E}\{u(Y, V, x) \mid D = d, X = x\}$. This formally includes the authors' approach in §3.7 if we take V to be the collection of all potential responses, although for the reasons outlined above I cannot support this choice. However, we might reasonably be able to make such an extension for an appropriate real-world quantity V—for instance, V might code for whether or not the defendant actually committed the crime. We would need additional data, in which V is actually observed, to estimate the required distribution of (Y, V) given (D, X), but as V is (unlike the collection of potential responses) a real-world quantity, there is no logical impediment to this.

I am grateful that this paper has got me thinking about these problems, and am pleased to second the vote of thanks to its authors.

Conflict of interest: None declared.

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The vote of thanks was passed by acclamation

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Andre Python's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment' by Imai et al.

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I congratulate the authors for providing a causal framework to evaluate how risk assessment scores generated by algorithms affect decisions taken by judges in the 'first appearance' hearing. Beyond direct applications in the judicial system, their work may inspire researchers in various fields to further improve the interface between algorithmic recommendations and the decision maker when humans are the ultimate decision maker. Their work contributes to address a fundamental question, *whether algorithmic recommendations help humans make better decisions*. This is particularly relevant in high-stakes decision making settings.

I believe that the methods suggested in the paper will find future applications in various research areas, including my own work focused on the quantitative study of political violence. In some contexts, interpretable algorithmic recommendations may help policy makers design and implement measures to prevent political violence (Python et al., 2021); however, an empirical assessment of the effects of such recommendations on policy makers is not always feasible. When decisions may result in e.g., the loss of human lives, assessing algorithmic recommendations via an RCT framework may not be desirable for ethical reasons. It would be helpful to discuss what alternative methods may be used to assess the relevance and impact of algorithmic recommendations on decisions when controlled experiments are not feasible or not desirable.

Furthermore, the PSA-DMF framework is deterministic and therefore sufficiently transparent and interpretable to ensure accountability, which is highly beneficial in this specific judicial setting. In other contexts, where the number of potential covariates is very large and/or their effects are not well understood, decision makers may be constrained to rely on more complex and less interpretable models. I am interested in further discussion on how the degree of interpretability of algorithmic recommendations may affect the decision maker, and more generally, how it may affect the *optimality* of the decision. As suggested by Rudin and Radin (2019), should we avoid using black-box algorithms for high stakes decisions if an interpretable model can achieve the same predictive performance? Also, given the growing capability of algorithms to make relevant recommendations, I wonder how an empirical evaluation of algorithmic recommendations may help determine who should be the ultimate decision maker, i.e., the man or the machine.

I wish to thank the authors for their valuable contribution and encourage the research community to continue to be engaged in this important interdisciplinary research area.

Conflicts of interest: None declared.

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Peng Ding's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment' by Imai et al.

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I want to congratulate the authors on a fascinating application of principal stratification (Frangakis & Rubin, 2002) and in particular, the principal score approach (Ding & Lu, 2017). I offer two technical comments.

First, I compare the classic instrumental variable (IV) model with the model of Imai et al. (2023). Figure 1 shows that both can be illustrated by similar causal diagrams with different definitions of the confounders between D and Y. Start with binary D and Y. The IV model (Angrist et al., 1996; Imbens & Angrist, 1994) assumes $D(1) \geq D(0)$, and the two-stage least-squares estimator (TSLS) identifies the complier average causal effect

$$E\{Y(1) - Y(0) \mid D(1) \geq D(0)\} = \frac{E\{Y(1) - Y(0)\}}{E\{D(1) - D(0)\}} = \frac{E(Y \mid Z = 1) - E(Y \mid Z = 0)}{E(D \mid Z = 1) - E(D \mid Z = 0)}.$$

In contrast, the model of Imai et al. (2023) assumes $Y(1) \leq Y(0)$ and ensures the sign-identifiability of the following principal causal effects (Imai et al., 2023, Theorem 1):

$$\begin{aligned}
&E\{D(1) - D(0) \mid Y(1) = 0,\ Y(0) = 1\} \propto E(Y \mid Z = 0) - E(Y \mid Z = 1),\\
&E\{D(1) - D(0) \mid Y(1) = 1,\ Y(0) = 1\} \propto E(DY \mid Z = 1) - E(DY \mid Z = 0),\\
&E\{D(1) - D(0) \mid Y(1) = 0,\ Y(0) = 0\} \propto E\{(1 - D)(1 - Y) \mid Z = 0\} - E\{(1 - D)(1 - Y) \mid Z = 1\}
\end{aligned}$$

up to positive, unidentifiable constants. The causal parameters conditional on $U_D = (D(1), D(0))$ and $U_Y = (Y(1), Y(0))$ have different scientific meanings.

Now consider an ordinal $D \in \{0, 1, \ldots, k\}$ and binary Y. The IV model (Angrist & Imbens, 1995) assumes $D(1) \geq D(0)$, and the TSLS estimator identifies a weighted average of causal effects for potentially overlapping subgroups (Angrist & Imbens, 1995, Theorem 1):

$$\frac{E(Y \mid Z = 1) - E(Y \mid Z = 0)}{E(D \mid Z = 1) - E(D \mid Z = 0)} = \sum_{j=1}^{k} w_j \cdot E\{Y(j) - Y(j-1) \mid D(1) \geq j > D(0)\}$$

with weights

$$w_j = \frac{P\{D(1) \geq j > D(0)\}}{\sum_{j'=1}^{k} P\{D(1) \geq j' > D(0)\}}, \quad (j = 1, \ldots, k).$$

In contrast, the model of Imai et al. (2023) assumes $Y(d_1) \leq Y(d_2)$ for $d_1 \geq d_2$ and simplifies the principal stratum $\{Y(d) : d = 0, 1, \ldots, k\}$ to an ordinal measure of risk:

$$R = \sum_{d=0}^{k} Y(d) \in \{0, 1, \ldots, k + 1\}$$

Interesting causal effects include

$$P\{D(1) = r \mid R = r\} - P\{D(0) = r \mid R = r\},$$

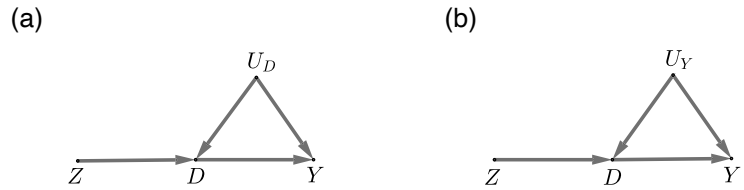

Figure 1. Causal diagrams with exclusion restriction but different 'confounders' between D and Y: (a) for the classic instrumental variable model (Angrist et al., 1996; Imbens & Angrist, 1994) with $U_D = (D(1), D(0))$, and (b) for the model of Imai et al. (2023) with $U_Y = (Y(1), Y(0))$ for binary D and $U_Y = \{Y(d) : d = 0, 1, \ldots, k\}$ for ordinal D.

which is not identifiable due to the complex principal strata. Imai et al. (2023) make additional strong and untestable assumptions for statistical inference.

Moreover, the IV model (Angrist & Imbens, 1995; Angrist et al., 1996; Imbens & Angrist, 1994) allows for general Y, whereas the model of Imai et al. (2023) requires binary Y. In principle, the latter can also deal with general Y, but it is a non-trivial extension.

Second, I extend Imai et al. (2023)'s discussion to a general causal estimand and propose an estimation strategy without imposing additional assumptions. With pre-specified $C_r(d_1, d_0)$, the estimand

$$\tau_r = \sum_{d_1, d_0} C_r(d_1, d_0) \cdot P\{D(1) = d_1, D(0) = d_0 \mid R = r\}$$

defines a general principal causal effect, which can depend on the joint distribution of $(D(1), D(0))$ as previously discussed by Lu et al. (2018, 2020) for ordinal variables. It includes the estimands in Imai et al. (2023) as special cases with appropriately chosen $C_r(d_1, d_0)$. In addition, other estimands can provide more detailed and relevant quantification of the effect of PSA provision on judges' decisions. I give two examples. The estimands

$$\begin{aligned} &P\{D(1) \geq r, D(0) < r \mid R = r\} \\ &= \sum_{d_1, d_0} 1(d_1 \geq r, d_0 < r) \cdot P\{D(1) = d_1, D(0) = d_0 \mid R = r\}, \quad (r = 1, \ldots, k) \end{aligned}$$

and

$$P\{D(1) = 0, D(0) > 0 \mid R = 0\} = \sum_{d_0} 1(d_0 > 0) \cdot P\{D(1) = 0, D(0) = d_0 \mid R = 0\}$$

can quantify the proportions of judge decisions positively affected by PSA provision given the risk measures of the cases.

Unfortunately, τ_r is not identifiable based on the observed data. Without imposing additional assumptions, we can only obtain its large-sample bounds. The proposed strategy hinges on the idea of classifying units into $(k+1)^2 \times (k+2)$ latent strata defined by $(D(0), D(0), R)$, which appeared before in causal inference (e.g., Balke & Pearl, 1997; Kline & Tartari, 2016; Kuroki et al., 2010; Mattei & Mealli, 2011; Yang & Ding, 2018; Yang & Small, 2016).

Define $p_{d_1, d_0, r} = P\{D(1) = d_1, D(0) = d_0, R = r\}$ as the probability of the latent strata $(D(0), D(0), R) = (d_1, d_0, r) \in \{0, 1, \ldots, k\} \times \{0, 1, \ldots, k\} \times \{0, 1, \ldots, k+1\}$. Vectorize them as $\boldsymbol{p}$. Table 1 shows the strata with binary D. Then τ_r can be written as the ratio of two linear functions of $\boldsymbol{p}$:

$$\tau_r = \frac{\sum_{d_1, d_0} C_r(d_1, d_0) \cdot P\{D(1) = d_1, D(0) = d_0, R = r\}}{P(R = r)} = \frac{\sum_{d_1, d_0} C_r(d_1, d_0) \cdot p_{d_1, d_0, r}}{\sum_{d_1, d_0} p_{d_1, d_0, r}}.$$

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Table 1. $4 \times 3 = 12$ latent groups defined by ($D(0)$, $D(0)$, $Y(0)$, $Y(1)$) or equivalently, ($D(0)$, $D(0)$, R) under the model of Imai et al. (2023) with binary Z, D, and Y

$D(0)$	$D(1)$	$Y(0)$	$Y(1)$	$R = Y(0) + Y(1)$
1	1	1	1	2
1	1	1	0	1
1	1	0	0	0
1	0	1	1	2
1	0	1	0	1
1	0	0	0	0
0	1	1	1	2
0	1	1	0	1
0	1	0	0	0
0	0	1	1	2
0	0	1	0	1
0	0	0	0	0

The observed data impose linear constraints on $\boldsymbol{p}$: for $d = 0, 1, \ldots, k$,

$$P(D = d, Y = 1 \mid Z = 1) = P\{D(1) = d, Y(d) = 1\} = \sum_{d_0} \sum_{r > d} p_{d,d_0,r},$$

$$P(D = d, Y = 0 \mid Z = 1) = P\{D(1) = d, Y(d) = 0\} = \sum_{d_0} \sum_{r \leq d} p_{d,d_0,r},$$

$$P(D = d, Y = 1 \mid Z = 0) = P\{D(0) = d, Y(d) = 1\} = \sum_{d_1} \sum_{r > d} p_{d_1,d,r},$$

$$P(D = d, Y = 0 \mid Z = 0) = P\{D(0) = d, Y(d) = 0\} = \sum_{d_1} \sum_{r \leq d} p_{d_1,d,r}.$$

Therefore, bounding τ_r is a linear-fractional programming problem, which can be reduced to a linear programming problem.

Conflict of interest: None declared.

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Pier Luigi Conti's contribution to the Discussion of 'Experimental Evaluation of Algorithm-Assisted Human Decision-Making: Application to Pretrial Public Safety Assessment' by Imai et al.

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First of all, the Authors deserve my warmest compliments for their nice piece of work. Below, my main discussion points.

The data used by Authors come from Dane county, Wisconsin. However, the Authors main interest is to give a general value to their conclusion, not simply confined to Dane county. Of course, extending conclusions outside of the small domain of Dane county is possible if Dane county data could be considered as if they were draw by simple random sampling from a larger population. More generally, this conclusion holds if data at hand could be considered as drawn by an *ignorable* sampling design (Pfeffermann, 1993), so that a purely model-based inference is adequate. In my opinion, the Authors should assess whether such an assumption defendable in the present case.

The above remark call for the notion of *representativeness* of the sample. Roughly speaking, a representative sample is a scaled-down version of the population, capturing its characteristics. Is the Dane county sample "representative" of a large population? Of course, the notion of "representative sample" is vague. As good starting point is the definition given in Grafström and Schelin (2014).

A possible way to widen the conclusion of the paper could consist in taking different counties, each one considered as a sample. The combination of sample data, in this case, is essentially a

problem of data integration, where data coming from different (possibly independent) samples should be integrated in order to draw inferential conclusions. It would be of interest to study which kind of assumptions we need in order to use integrated data for evaluating causa effects, as well as how difficult is the extension of Authors' analysis to this more general setting.

The optimal decision rule developed by the authors depends on two parameters, c_0 (cost of FTA, or NCA or NVCA) and c_1 (cost of unnecessary harsh decision). Are there any indication for c_0, c_1? Furthermore, it is essentially thought as an "optimal recommendation" for the judge. It would be of interest to assess whether such a rule optimizes the judge's final decision. This remark calls for a related one. Is the judge's decision algorithmically predictable? And, after all, should the algorithm assisting the judge mimic the judge decision process? In a sense, such an algorithm should mimic an "optimal judge," if this is of any meaning.

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Aaron Sarvet and Mats Stensrud's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment' by Imai et al., part 1

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> *He who understands [my propositions] finally recognizes them as senseless, when he has climbed out through them, on them, over them…must so to speak throw away the ladder, after he has climbed up on it.*
>
> Wittgenstein (1922), *Tractatus logico-philosophicus*

Principal stratum estimands were once disparagingly compared to a 'magic can opener' (Dawid & Didelez, 2012)—an allusion to a classic joke made on behalf of mathematicians and economists for their perceived tendency to assume into existence solutions to practical problems. The implication was that the use of (joint) potential outcomes often lead to impractical and logically unverifiable solutions constructed on imaginary grounds.

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Many will find the ontological implications of a potential outcomes framework suspect or even disturbing. These concerns are highlighted in Imai et al.'s (IJGHS's) juridical context: we are asked to consider strata of individuals defined by what crimes they will (or will not commit), depending on a judge's decision. We must consider the so-called 'risky-cases' ($Y^{d=1} = Y^{d=0} = 1$): guilty from the day they are born, the 'risky-cases' are fated to commit a crime regardless of a judge's decision. Furthermore, principal strata are completely unmeasurable, and yet we are asked to consider strong conditions on their distribution. IJGHS ask us to believe that there could be no individuals with ($Y^{d=1} = 1$, $Y^{d=0} = 0$) whenever their methodology would be used (monotonicity).

Nevertheless, it is hard to deny that potential outcomes, and related counterfactual frameworks, have proved useful as reasoning tools (see e.g., Richardson & Robins, 2013). Thus, they might be more aptly considered as a 'ladder' than an 'imaginary can-opener' (Robins et al., 2020). But this does not mean they are always *necessary*—in many cases, analogous estimands are identifiable using the exact same functionals of observed data parameters, except under models that are agnostic to the existence of potential outcomes (Dawid, 2021). These agnostic formulations allows us to 'throw away the ladder' if we so please—and in circumstances that consider crime and punishment, likely we should.

Unfortunately, no such agnostic conceptual equivalents to the estimands of IJGHS exist. Perhaps most illustratively, IJGHS suggest a definition of an optimal decision rule in which investigator-specified costs are assigned to average treatment effects conditional on principal strata. They further suggest 'policy makers could [...] adopt this rule as the recommendation for judges'. But there is no possible experiment that could verify inferences on this parameter—in other words, there do not exist any set of sufficient identification assumptions which hold by design in any real-world study.

IJGHS climb the ladder of potential outcomes to reach their solution. But we are asked to simply (and forever) have faith in its assumptions; and our ladder: we cannot throw it away. As such, we might consider Wittgenstein's later advice:

> *If the place I want to get could only be reached by way of a ladder, I would give up trying to get there.*
> Wittgenstein (1977), *Culture and value*

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Mats Stensrud, Vanessa Didelez and Aaron Sarvet's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment' by Imai et al., part 2

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Imai et al. (IJGHS) have conducted a timely experiment on evaluating a decision support algorithm. However, we are concerned by their choice of estimands which, even if they appear plausible at first, rely on notions and assumptions for which we cannot ever obtain empirical evidence. We suggest an alternative.

We adopt the notation of IJGHS, but use superscripts for potential outcomes. To simplify, we consider a binary decision and outcome ($Y, D \in \{0, 1\}$).

1 Alternative single-world estimands

With the randomisation of IJGHS's experiment, we can identify *without further assumptions* the following single-world outcome parameters (SWOPs) for $z \in \{0, 1\}$:

- $P(Y^z = 1, D^z = 1)$ failed detention (crime and detained);
- $P(Y^z = 0, D^z = 0)$ successful releases (no crime and released);
- $P(Y^z = 1, D^z = 0)$ crime and released; and
- $P(Y^z = 0, D^z = 1)$ no crime and detained.

To evaluate whether 'algorithms help humans make better decisions', we could assign utility to each SWOP and find the regime that maximises this utility. Suppose practitioners (qualitatively) find it desirable that using an algorithm that:

1. reduces the marginal number of crimes,
   $P(Y^z = 1) = P(Y^z = 1, D^z = 1) + P(Y^z = 1, D^z = 0)$,
2. reduces the number of failed detentions,
   $P(Y^z = 1, D^z = 1)$, and
3. increases the number of successful releases,
   $P(Y^z = 0, D^z = 0)$.

IJGHS main results, Theorem 1, allows qualitative (sign) identification of APCEs under the exclusion restriction (Assumption 2) and the (unverifiable) monotonicity assumption (Assumption 3). When these conditions hold, however, our practitioners will always come to the same qualitative

conclusion as IJGHS because

$$\begin{aligned}
\text{APCE}_p &= \alpha\{[P(Y^{z=1}=1, D^{z=1}=1) + P(Y^{z=1}=1, D^{z=1}=0)] \\
&\quad - [P(Y^{z=0}=1, D^{z=0}=1) + P(Y^{z=0}=1, D^{z=0}=0)]\}, \\
\text{APCE}_r &= \beta\{P(Y^{z=1}=1, D^{z=1}=1) - P(Y^{z=0}=1, D^{z=0}=1)\}, \\
\text{and} \quad \text{APCE}_s &= \gamma\{P(Y^{z=1}=0, D^{z=1}=0) - P(Y^{z=0}=0, D^{z=0}=0)\},
\end{aligned}$$

where $\alpha = P(Y^{d=0}=1) - P(Y^{d=1}=1)$, $\beta = P(Y^{d=1}=1)$, and $\gamma = P(Y^{d=0}=1)$.

Crucially, in contrast to IJGHS, optimising a utility over the SWOPs will yield an interpretable treatment rule regardless of whether monotonicity or other unverifiable assumptions hold.

Alternatively, when monotonicity fails, e.g., if individuals engage in NCA only if detained (because prison is a more aggressive environment), how biased would IJGHS' approach be?

2 Optimal decisions

IJGHS specify two APCE parameters which quantify utility of counterfactual outcomes that are unobservable and unverifiable, even in principle. The motivation for these utilities, however, is to quantify

- 'an NCA (NVCA or FTA)' (c_0), and
- 'an unnecessarily harsh decision' (c_1).

Similar to our above argument, consider practitioners who assign marginal utilities

- $c_0' = 1 + c_0$ for the cost of a crime ($Y = 1$) regardless of D, and
- $c_1' = c_1$ for releasing an individual who does not commit a crime, ($Y = 0$, $D = 0$).

Whenever IJGHS's Assumptions 1–4 hold, our practitioners will come to exactly the same conclusion as IJGHS about 'optimal' regimes, but our practitioners' conclusion will still be meaningful without those assumptions.

3 Principal fairness

We doubt that IJGHS's principal fairness criterion adequately captures notions of fairness, which involve much more than an independence condition. The comment by Sarvet & Stensrud discusses the unappealing philosophical properties of this approach.

Alternatively, even if we accepted 'principal fairness' as a suitable criterion, IJGHS are (internally) inconsistent: IJGHS use principal fairness marginally, but Imai and Jiang (2020) explicitly stated the importance of evaluating principal fairness conditional on covariates. This inconsistency is a serious problem, because conditional independence does not imply marginal independence and vice versa.

Conflict of interest: None declared.

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Efrén Cruz Cortés and Debashis Ghosh's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human-decision making: application to pretrial public safety assessment' by Imai et al.

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We commend the authors on a thought-provoking paper (Imai et al., 2022) combining real-life experiments and a new causal inferential framework for evaluating fairness. One hidden strength of the paper is the direction it takes regarding ethical artificial intelligence (AI) research. Performance of algorithms in terms of 'fairness' metrics has received much focus. However, enforcing these metrics does not necessarily rid the system under consideration of structural harm and discrimination (Cortés & Ghosh, 2020; Knox et al., 2020), partly due to the social components of the system which still exist and contribute to the generation and automation of harms. In this case, the social component is the algorithmic influence on human decision makers. By shifting the point of focus from algorithmic performance to algorithmic influence on social systems, we move from the detached, possibly ineffectual properties of technical objects, to the holistic, emergent properties of technosociety (Cortés et al., 2021; Ferguson, 2017; Selbst et al., 2019).

As an academic community we grapple with the question: what makes AI decisions different to human decisions? As we know, bias and discrimination are socially produced phenomena antecedent to automation (Bowker & Leigh Star, 2000), a history reflected in human-generated data patterns reproduced by algorithms (Benjamin, 2019; Crawford, 2021; Eubanks, 2017; O'Neil, 2016). Is automated decision making, then, fundamentally different to human decision making? Or is it the same but bigger and faster? As simple as this question is, we do not have a conclusive answer yet. An intermediate answer exercises a similar analysis as in this paper, exploring instead how automation influences already rooted human decision-making protocols.

The effects measured in the paper concern the presence or the absence of algorithmic influence on judges. To push further the identification of generating points of harm in human–computer systems, we suggest some guiding questions: (1) Are algorithms more or less lenient than judges for the different case divisions (division over race and division over gender)? (2) What exactly is the mechanism driving judges to take more or less lenient decisions? (3) When judges are influenced, are the judges' responses positively correlated to algorithmic recommendations? Or conversely, are the judges becoming stricter because they think they are counteracting a lenient algorithm? (4) Why are we not seeing those effects in some cases? (5) When we do not observe influence, is it because the algorithm confirms biases the judge already has, or because the judge exhibits such strong bias that no matter what the algorithm recommends the judge will proceed in their standard manner? Answering these fundamental questions may require challenging some consistency assumptions (Rehkopf et al., 2016), it will also require both qualitative and quantitative research.

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Data availability

No new data were generated or analysed in support of this research.

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Eugenie Hunsicker's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment' by Imai et al.

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A judge's decision about bail influences if someone remains in custody or not until trial, but the relationship is not direct—socioeconomic factors also contribute enormously. Have the authors considered how the variable 'Remained in custody'/'Did not remain in custody' mediates the

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relationship between bail decision and the variable 'Y', fails to appear, commits additional crime or violent crime.

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Jane Luise Hutton's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment' by Imai et al.

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I read this paper from a background in randomised controlled clinical trials, and I congratulate the authors on achieving randomised research in a legal system (Fernandez Lynch et al., 2020). Given the focus on judges' decisions, I was puzzled that it was simply an individually randomised trial. I now appreciate that discussion of optimal cluster randomized designs is a future project. Deciding how to translate analyses which model parameters for judges, and perhaps the sessions and courts, as well as characteristics of arrestees, for a legal audience will be challenging. As an aside, I wonder whether a parameteric analysis would provide more precise estimates and clearer communication. You might find the CONSORT Consort—Cluster Trials (consort-statement.org) and cluster trials CONSORT statements useful in persuading lawyers to use the appropriate designs. The literature on evaluating complex health care interventions might also be relevant in considering wider judicial systems. New framework on complex interventions to improve health—UKRI. (https://www.ukri.org/news/new-framework-on-complex-interventions-to-improve-health/)

Questions were asked about the ethics, and the role of consent in this study. I have dissected the structural issues relevant to the ethics of cluster randomised designs which distinguish between the units of randomisation, intervention and data collection (Hutton, 2001). Consent, informed or not, is not the only ethical issue (Hutton, 2001; Hutton & Ashcroft, 2000). Your work seems similar to what is called implementation research in health care. I have argued that the appropriate ethical principles are those of professions. The profession (lawyers) has a duty to execute good research (Hutton et al., 2009). Individual professionals have a duty to continue their professional education, which might include participation in generating knowledge.

I appreciated your further careful discussion of the definition of fairness. Fairness is not one dimensional, and I suspect the minimal satisfactory statistic is two dimensional, expressing risk to the public and penalties for individuals. This implies allowing separate and explicit utilities. In the medical context, racialised decision making can be critical: for example, in interpreting haemoglobin test results. Sex is also often important in diagnosis and treatment. It is far from clear to me

that the risks associated with men and women, particularly in relation to violent crime, should not be considered. Sex is not included in PSA, but I think careers in violent crime differ by sex. Ignoring differences is not better than understanding differences. 'There is nothing so unequal as the equal treatment of unequals.'

Conflicts of interest: None declared.

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Victor Richmond Jose's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment' by Imai et al.

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I congratulate the authors for a thought-provoking and timely paper related to justice, equity, and diversity. Given the space constraints, I would like to focus my comments focus on the decision-making framework (DMF) often neglected in applications. I also caveat that my comments below are more geared towards extensions and possible future work stemming from this pioneering RCT.

The two specific comments I have are about the utility form and fairness analysis under the DMF. Though a simple step function utility form is a good starting point, I envision future implementations would go beyond a misclassification-type cost structure $U(\delta)$. Having a form that takes only three possible values reduces the optimal decision to setting the expected proportion for three groups (cf. Eq. (13)). In practice, costs will likely be dependent on $\boldsymbol{X}_i$ as well as the deviation $\delta(\boldsymbol{X}_i) - R_i$. This means that misclassified individuals in one direction would not be all identical and more nuanced DMF results beyond the relative comparison of the cost ratio c_1/c_0 (implied by the plots and discussion in S10) will emerge.

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Regarding the analysis of gender and racial fairness, the algorithm did not show any egregious fairness concerns. This outcome however is not always guaranteed. I believe that the next step should move away from an *ex post* analysis of fairness to a more deliberate choice of promoting fairness in a DMF. One could perhaps make fairness integral to the objective or utility function. For example, a multi-objective utility function that includes fairness can be used and will force decision makers to reflect about the tradeoffs between potentially conflicting objectives. Alternatively, fairness constraints could be used. Though there is no legal definition or consensus of what fairness means, recent legal applications have used benchmarks like the 'four-fifths rule' (Cascio & Aguinis, 2001). Something similar could potentially be used here (Shimao et al., 2022).

In both suggestions, the use of discriminatory-prone variables is essential. Explicitly using them in statistical models might be contentious and challenging. Recent works acknowledging this challenge while reducing bias have started emerging (Aliverti et al., 2021; Pope & Sydnor, 2011). I think these pre-processing or post-processing methods could be well-integrated with the paper's DMF because of its generality.

I thank again the authors for this important contribution and look forward to seeing the full results of the 30-month study. I am hopeful more models such as this are implemented in practice.

Conflicts of interest: None declared.

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Kuldeep Kumar's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human-decision-making: application to pretrial public safety assessment' by Imai et al.

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I must congratulate the authors for developing the methodology for evaluating the causal impacts of algorithmic recommendations on human decisions and conducting a randomized controlled trial to prove its effectiveness in the area of the criminal justice system. I have no comments on the proposed evaluation methodology, which looks sound to me. Still, I am not sure about the practical applications in the criminal justice system and other areas.

I am not sure how the Public Safety Assessment (PSA) score is calculated, but it cannot consider many risk factors that the human mind can take into account based on experience, knowledge, skills and patterns in the past. This is what is called 'Tacit Knowledge' and lots of work has been done in the area of converting tacit knowledge into 'Explicit Knowledge'. While making a judgement based on tacit knowledge, judges can always make Type I and Type II errors. In this context, it is difficult to assess the misclassification costs (releasing the defendant when the defendant is guilty vis a vis, not releasing the defendant when the defendant is not guilty). It will be interesting to compare Type I and Type II error based on tacit knowledge of the judges and based on explicit knowledge like PSA or risk assessment score. Overall, I always believe in the hybrid model, and in the present scenario, judges can have their judgement based on the past pattern and then take PSA Score as well if the judgement is other way round. However, it may violate Assumption 2 as the PSA score may directly influence the judge's decision.

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Tyler VanderWeele's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment' by Imai et al.

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Imai et al. (2023) are to be commended for their thoughtful evaluation of the Public Safety Assessment (PSA) in judicial decision-making. Rigorous designs for evaluating how predictive algorithms can assist human decision-making are very much needed. While deep learning algorithms have advanced considerably, in many contexts it is important not to abandon human decision-making. Such contexts include settings in which there are special considerations which are not easily coded (e.g., an experienced physician's intuition; or specific details of a criminal case), settings in which there are incommensurable ends so that it is not possible to specify a univariate objective function (e.g., admissions trade-offs between diversity and academic preparedness), settings in which moral considerations are not easily embedded as constraints, and settings in which human decision-making or autonomy is itself partially constitutive of human flourishing (e.g., choosing one's spouse). However, even in such settings predictive algorithms can be of assistance in human decision-making. They can provide additional information or estimates (difficult for the human mind to carry out) that can be used, amongst other considerations, in decision-making. They can help identify and eliminate choices that are clearly suboptimal when competing ends are in play. They can help identify a set of the most promising possibilities when the options are vast and yet human choice is itself important (e.g., online-dating match-algorithms).

Using an approximately randomised design, Imai et al. (2023) evaluate the effects of providing PSA information on judicial decision-making, and on the fairness of those decisions. However, an important issue in all algorithm-assisted human decision-making contexts concerns exactly what information to provide, i.e., what information will most aid decision-makers? This point is not discussed by Imai et al. (2023), perhaps reasonably, because their purpose is to evaluate the PSA information and system already in place. However, other information may have been more useful to judges. For example, one could imagine presenting instead the actual predicted probabilities of 'failure to appear' and 'new criminal activity' (rather than the somewhat convoluted and difficult-to-interpret PSA). Elsewhere, I have proposed such an approach in medical decision-making in which clinicians are provided predicted outcome-probabilities for all treatment options but then make, in conjunction with the patient, the actual decision, accompanied by randomised trials to evaluate the effects of providing this additional information (VanderWeele et al., 2019; cf. Wu et al., 2020). Such predicted probabilities might well have been more valuable to the judges than the PSA. It is also conceivable that randomised trials could evaluate the effects of providing this information versus PSA. Again, further work on optimising algorithms for human decision-making contexts, on evaluating these algorithms, and on choosing which information to present, will be important in future research.

Conflicts of interest: none declared.

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Priyantha Wijayatunga's contribution to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to the pretrial safety assessment' by Imai et al.

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I would like to make few points as far as building a general model for contexts such as that of the authors'. Even though the potential outcome framework is very popular among many statisticians and empirical analysts for different types of causal inferences in many subject domains, so-called graphical modelling (GM) (Pearl, 2009) approach is appealing since it makes the context of the problem clearer to the reader, in addition to its ability to facilitate communication between the statistical modeller and the subject domain expert. In modelling, it is important to take into account all the factors that relate to the outcome of interest in one way or another, if one wants to get the more general inferences. Perhaps not all of them have direct effects on the outcome, especially when some of them become conditionally independent of the outcome when some of the others are known, etc. Such relationships can be shown and modelled effectively with the GM framework. Note that the interaction between the statistical modeller and the subject domain expert is important since, the former can have an impact on the latter even to expose details of the modelling context, that might not have exposed initially but are important. Another advantage of the GM approach is that one can use effectively expert opinion of the context, for example, the opinion that can be encoded as hypothetical data counts, through Bayesian parameterisation of the models. This is extremely useful in cases where the observed data are not sufficient and also to generalise the model inferences to a broader scope than just the current context. The latter is about external validity of the inferences. Furthermore, since the theory of the GM is well established by now, an applied analysis based on it may not contain any new theoretical results, which is pleasing to many applied users. Though the theoretical results are important, in order to

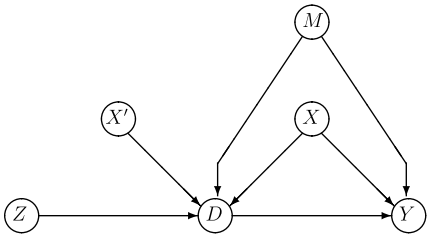

Figure 1. A possible causal structure for the experiment.

popularise statistical methodologies we need more works that only contain the application of the theoretical results, ideally discussing practicality of them.

In the application of this paper, it may be worthwhile to mention about pretreatment covariates of judges, say, X' that affects their decisions and macro-socio-economic factors (M) that may affect the judges' decisions and also the different types of crimes. However, unlike in authors' context, then we need a lot of data across different macro-socio-economic contexts. In the event of availability of such data, we can, for example, calculate the causal effect of Z on Y by calculating the intervention probability (Figure 1)

$$\begin{aligned} p(y \mid do(z)) &= \sum_{x,x',m,d} p(x)p(x')p(m)p(d \mid x, x', m, z)p(y \mid x, m, d) \\ &= \sum_{x,x',m,d} p(x)p(x')p(m)p(d \mid x, x', m, z)p(y \mid x, x', m, z, d) \\ &= \sum_{x,x',m} p(x)p(x')p(m)p(y \mid x, x', m, z). \end{aligned}$$

When the model has been built as an inference system, it can be applied across many similar but distinct scenarios. In the authors' case, it should be made clear that on which macro socio-economic contexts, etc., the inferences are obtained and valid.

Conflict of interest: The author has no any conflict or interest.

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The authors replied later, in writing, as follows

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Authors' reply to the Discussion of 'Experimental evaluation of algorithm-assisted human decision-making: application to pretrial public safety assessment.'

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We thank the Discussion Meetings Committee of the Royal Statistical Society for providing us with a valuable opportunity to present our paper and receive feedback from many scholars. Despite their diverse perspectives, we believe that all of our discussants are in agreement about the need to develop new statistical methodology to better analyze algorithm-assisted human decision-making.

While the use of algorithms has become ubiquitous in today's society, we—humans—still make many consequential decisions with the help of algorithms. As Kumar and VanderWeele noted, such a hybrid decision-making system may allows one to combine human experiences and knowledge with algorithmic recommendations, possibly leading to improved decisions. Moreover, even if fully algorithmic decision-making is more optimal than a hybrid system, we may still prefer the latter because we want to hold humans, rather than algorithms, accountable for the consequences of decisions.

The importance of algorithm-assisted human decision-making motivated us to develop a set of methodological tools to evaluate and understand how algorithmic recommendations affect human decisions. As echoed by Cruz Cortéz and Gosh, our study sharply contrasts with much of the existing studies whose focus has been the accuracy and fairness of algorithms themselves. We hope other researchers follow up on this important research agenda. Below, we respond to specific comments raised by the discussants.

Scientific and policy-relevant questions. When analyzing algorithm-assisted human decision-making, one important question is whether algorithmic recommendations help humans make better decisions. In the context of the pretrial public safety assessment (PSA), the goal of judge's decision is to prevent crimes while avoiding unnecessarily harsh decisions. If the PSA helps them achieve this goal, it should nudge judges to make a harsher decision in the case that would prevent an arrestee from committing a crime while encouraging them to make a more lenient decision for those who would not cause a negative outcome regardless of judge's decision. Our causal estimands (Average Principal Causal Effects or APCEs) directly addresses this key question of whether and how algorithmic recommendations help human decision makers achieve their goal. Our methodology shows how to learn about these causal quantities from the observed data.

Some discussants such as Dawid argue that the simple comparison of human decisions (and the downstream outcomes) with and without algorithmic recommendations can answer this question. In the case of our randomized controlled trial, they suggest that we estimate the causal effect of

PSA provision on decisions by judges and the negative outcomes (see Figure 2 of our article for the results of this analysis). Unfortunately, although such an intention-to-treat (ITT) analysis evaluates whether the provision of algorithmic recommendation affects judicial decisions and/or arrestees behavior, it cannot answer the essential question of whether PSA provision helps judges achieve their goal. Suppose, for example, that the ITT analysis shows the provision of PSA leads to harsher decisions and reduces the number of crimes. This evidence alone cannot rule out the possibility that the PSA provision leads to many unnecessarily harsh decisions. To properly evaluate the efficacy of PSA, we must understand whether it encourages judges to make a harsh decision only when it helps prevent a crime. This is exactly what our proposed methodology accomplishes.

More generally, as many discussants (e.g., Cuellar, Cruz Cortéz and Gosh, Ding, Hunsicke, Kumar, Python, VanderWeele, and Wijayatunga) recognize, the evaluation of algorithm-assisted human decision-making requires analyzing the *mechanisms*, through which algorithmic recommendations influence human decisions. Such an analysis is critical for policy makers who are considering the adoption of algorithmic recommendations. It is also essential for social scientists whose goal is to understand how humans incorporate algorithmic recommendations into their decisions. Simply estimating the overall ITT effects of algorithmic recommendation on human decisions and downstream outcomes is not sufficient for these purposes.

Essential role of potential outcomes and principal stratification. Some discussants dispute the value of our methodology and advocate an alternative approach. Dawid criticizes the use of principal stratification while noting that 'some of the proponents of these arguments have been honored with the 2021 Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel.' We let readers decide whether approaches based on potential outcomes and principal stratification, including ours, are useful or 'totally misguided.' Instead of engaging with this philosophical debate, we examine Dawid's recommendation to use the statistical decision theory for answering the following questions: (1) 'should we present PSA to the judge before she makes her decision, or not?', and (2) 'How, ideally, should the judge use the information she has in coming to her decision?'

Dawid is incorrect to assume that his first question can be answered by the ITT analysis alone. As explained above, to decide on the use of PSA, policy makers need to understand if PSA helps judges achieve their goal of preventing a negative outcome while avoiding unnecessarily harsh decisions. The ITT analysis alone cannot answer this key question of the PSA evaluation study.

Answering Dawid's second question also requires going beyond the analysis he suggests. As noted earlier, although a harsher decision typically reduces the probability of a negative outcome, a judge does not wish to issue the harshest decision to every arrestee. In fact, one goal of the PSA is to help a judge identify arrestees who would commit a crime only if they are given a lenient decision. According to this goal, an optimal decision needs to balance the cost of mistakenly detaining an arrestee who would not commit a crime regardless of the decision, against the cost of releasing an arrestee who would commit a crime only when released. Thus, joint potential outcomes are necessary to characterize the risk levels of arrestees.

In addition, these costs may be different. For example, the cost of mistakenly detaining 'safe' arrestees may exceed that of releasing 'risky' arrestees. In medicine, the Hippocratic principle to 'do no harm' has the same asymmetric utility structure: mistakenly killing a person is considered more costly than failing to save an individual. In Ben-Michael et al. (2022), we consider the optimal decision rule under such settings. Although the observed data do not identify the joint distribution of potential outcomes, we propose first to partially identify the expected utility and then to find the optimal policy in the worst scenario. Any thoughtful evaluation of algorithm-assisted human decision-making must directly address whether algorithmic recommendations help achieve the goal of human decision makers. Potential outcomes and principal stratification are useful tools for conducting such evaluation under various settings.

We briefly respond to Stensrud, Didelez, and Sarvet who make a similar criticism about the use of joint potential outcomes. They propose alternative 'single-world' estimands, including $\{Y_i(Z_i = z) = D_i(Z_i = z) = 1\}$ which they refer to as 'failed detention.' But, in what sense, does this event represent a failure? If an arrestee is someone who would commit a crime regardless of the decision, it is unclear if the decision to detain this person is a failure. Without consideration of counterfactual outcome, one cannot determine whether a correct decision is made for each case.

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Stensrud et al. further suggest that analysts focus on the same decision problems as the ones proposed by Dawid. They show that there exist direct relations between these decision problems and our causal quantities of interests. Although the existence of such relations is interesting, this does not alter the fact that without considering joint potential outcomes, one cannot even define causal quantities of interest related to the goal of judges—prevent negative outcomes while avoiding unnecessarily harsh decisions. More importantly, these relations exist under our identification strategies. In many causal analyses, it is useful to first define causal quantities of interest, using potential outcomes, and then consider how various assumptions can partially identify or point-identify them.

We disagree with Dawid, Didelez, Sarvet, and Stensrud who object to the use of principal stratification in general. In contrast, we believe that principal stratification and potential outcomes are a powerful language for causal inference. Many researchers studied causal quantities of interest that involve both potential outcomes, including individual treatment effects ($Y_i(1) - Y_i(0)$) (e.g., Lei & Candes., 2021), the proportion of those who were harmed by the treatment ($\Pr(Y_i(1) - Y_i(0) < 0)$) (e.g., Kallus, 2022), quantiles of treatment effects (e.g., Fan & Park., 2012), and probability of necessary and sufficient causes ($\Pr(Y_i(t) = y \mid Y_i(t') = y')$) (e.g., Pearl, 2000) to name a few.

As these examples and their scientific applications demonstrate, consideration of joint potential outcomes is an essential aspect of causal inference. Instead of engaging in philosophical debates, we strongly encourage these discussants to get directly involved in applied causal inference research through direct collaboration with policy makers and scientists. We believe that doing so will help them gain a deeper appreciation of the important roles played by potential outcomes and principal stratification in determining causal estimands and communicating scientific and policy-relevant findings.

Verifiability of identification assumptions. In response to the critiques by Stensrud, Didelez, and Sarvet, who highlight the unverifiability of our identification assumptions, we emphasize that both randomization of treatment assignment (Assumption 1) and exclusion restriction (Assumption 2) are guaranteed to satisfied under the study design. The monotonicity assumption (Assumption 3) is based on the domain knowledge (It is worth noting that many researchers in this literature assumes a stronger monotonicity condition that an arrestee cannot commit a crime if detained [e.g., Coston et al., 2020]). Under these three assumptions, we have shown that the sign of our causal quantity of interest is identified. Ding, in his discussion, further shows how to derive sharp partial identification bounds in more general settings.

Finally, unconfoundedness (Assumption 4) is also guaranteed to hold if analysts have access to all the information used by a judge when making the decision. We are currently in the process of collecting additional information including the transcripts of court proceedings. In our paper, we develop a sensitivity analysis that allows one to examine the robustness of empirical findings to possible violation of this assumption. In causal inference, partial identification bounds and sensitivity analysis play a crucial role in examining the significance of different identification assumptions.

Fairness and ethics. Some discussants raise the issues about fairness. In our paper, we used the concept of principal fairness, which was introduced in a separate paper (Imai & Jiang., 2022). The paper is forthcoming in *Statistical Science*. Cuellar points out a large literature on fairness in computer science and statistics and wonders why we used principal fairness. We briefly respond here while leaving a more detailed discussion to the aforementioned paper. The main advantage of principal fairness is that it directly takes into account how the decision affects individuals. This consideration is critical in our study because judges' decisions can affect individual arrestees. In contrast, most existing fairness definitions, which are based on observed or single potential outcome, are unable to incorporate the causal effect of decision on the outcome of interest.

In addition, Hutton suggests that when considering fairness, risks associated with men and women should be considered separately. Stensrud, Didelez, and Sarvet point out that while an earlier version of our *Statistical Science* paper considered conditional principal fairness, our analysis focuses on the marginal principal fairness. The question of what variables to condition on is not statistical. Rather, it is an important question that confronts policy makers and the society. Finally, Jose suggests that fairness should be incorporated either in the utility or in the constraint when considering optimal decisions. We have written a related paper that incorporates principal strata in policy learning (Ben-Michael et al., 2022). This framework can directly incorporate the principal fairness criteria into policy learning.

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Finally, several commentators asked why our study design did not involve blinding of judges either to the intervention or to the fact of the study. First, it was not possible to keep the judges unaware of the presence of the risk assessment instrument in some cases but not others. Second, according to ethical principles, if an individual's decision-making is the focus of a study, researchers should, whenever possible, obtain their consent to participate in the study. In this context, eliciting informed consent from the judges requires informing them about the study.

Future research directions. There are several avenues for future research. The first is the question of how to improve the way in which the PSA is calculated. The key challenge lies in the fact that like many algorithmic recommendations used in public policy, the PSA is a deterministic rule. This means that learning a better rule from the observed data requires some degree of extrapolation. In Ben-Michael et al. (2021), we use the minimax statistical decision theory to limit the probability that the learned new rule performs worse than the existing rule. This is achieved by first specifying a model class for the conditional average treatment effect, which partially identifies the counterfactual outcome, and then maximizing the expected utility in the worst case using robust optimization. Another possibility is to present the algorithmic recommendations to judges in a different way. For example, VanderWeele suggests that the use of predicted probabilities, rather than integer values as done in the PSA, might better convey the relevant information to human decision makers.

Second, the proposed methodology can be extended to reflect additional real-world complexities. For example, judges make decisions about monitoring conditions as well as bail amount. We can also analyze multiple outcomes jointly rather than one outcome at a time as done in the current paper. Such generalization of our methodological framework will allow us to provide a more holistic evaluation of the PSA. For example, Ding generalizes our partial identification result by showing how to compute sharp bounds under the ordinal decision without additional assumptions. One may apply a similar analytical strategy to even more general settings such as multidimensional decisions and outcomes.

Third, although we have focused on the analysis of first arrests in our article, a relatively large number of people are rearrested in our data. Developing statistical methods for dealing with multiple arrests is important because it enables the use of all available data and answers important questions related to recidivism and accumulated impact of PSA on judges' decisions and arrestees' behavior.

Finally, Conti raises an important issue about the external validity while Cuellar emphasizes the value of conducting additional experiments with more judges and jurisdictions. In particular, they question how generalizable our empirical findings are given that the data come from Dane county, Wisconsin alone. We are currently conducting multiple experiments in other jurisdictions and hope to tackle the challenge of developing new methods to learn from multiple experiments in different sites.

Conflict of interest: None declared.

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